

6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

M. Tolentino^{1,2}, J.A. Bravo-Aranda^{1,2}, M.J. Granados-Muñoz^{1,2}, J.L. Guerrero-Rascado^{1,2}, and L. Alados-Arboledas^{1,2}

¹Department of Applied Physics, University of Granada, Granada, 18072, Spain

²Andalusian Institute for Earth System Research, Granada (IISTA-CEAMA), 18006, Spain

Correspondence: M. Tolentino (mtolentino@ugr.es)

Copyright statement. TEXT

1 Experimental site and Instrumentation

An extensive dataset spanning the years 2018 to 2023 was utilized in this study. Specifically, high-level post-processed data from Cloudnet (?) such as categorize, target classification, and microphysical files, along with humidity and temperature profiles generated by the microwave radiometer, were employed. Figure 1 illustrates the monthly Cloudnet high level products available for the AGORA ACTRIS-CCRES station (37°2'E, 3°6'S), at 680 m of altitude. The availability of these data depends on simultaneous vertically pointing measurements of the microwave radiometer, ceilometer, Doppler cloud radar, as well as the meteorological model products - the ECMWF model (acronym for European Centre for Medium-Range Weather Forecasts). This figure shows that data collection efforts improved over time, with 2020 exhibiting the highest and most consistent data availability throughout the year. In contrast, 2018 had no data for the early months, indicating that data collection commenced mid-year. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements. Despite these challenges, these three instruments have been continuously operated since 2018.

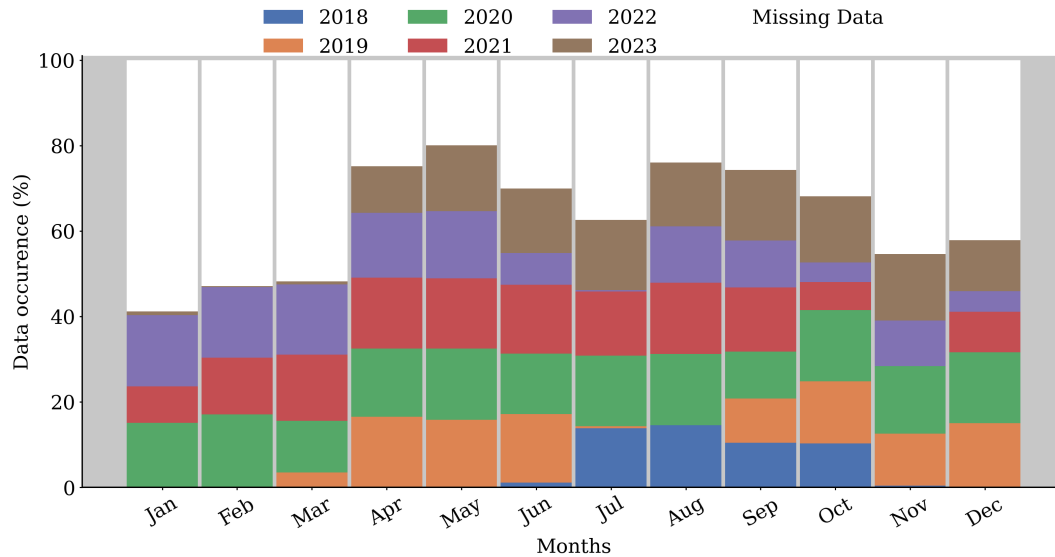


Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represent the data availability for different years.

It is worthy to note that this dataset represents a robust database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. The ACTRIS-CCRES Granada site is uniquely positioned as one of the few locations dedicated to cloud studies in a warm region that is frequently affected by synoptic conditions, Sahara dust intrusion, and thermodynamic conditions. Marinou et al.; F. J. Bello-Millán et al.; Casquero-Vera et al.. This dataset contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas.

1.1 Cloudnet database

1.1.1 Instruments

The Cloudnet target classification utilizes the remote sensing state-of-art instruments to study cloud properties and atmospheric phenomena. At Granada station, since 2010, a RPG-HATPRO G2 microwave radiometer has been employed, featuring 11 channels around the water vapor absorption band (23.8 GHz), 9 channels around the oxygen absorption band (52 GHz), and 10 channels around the liquid water absorption band (31.4 GHz). This instrument measures brightness temperature with a resolution of 0.3-0.4 K, which is subsequently used to calculate the liquid water path (LWP) and derive the liquid water content (LWC) for liquid clouds. Complementing this, since 2015, a CHM15k Nimbus ceilometer operating at a power of 8.4 μ J and wavelength of 1064 nm has been used to provide the attenuated backscattering coefficient of aerosols and cloud droplets. This instrument is used to estimating cloud base height and detecting aerosol layers due to the sensitivity of this wavelength to small

particles such as aerosols and water droplets. Additionally, a dual polarimetric RPG-FMCW 94 GHz Doppler cloud radar were installed at the our station in 2018. It measures the Doppler velocity spectrum (DVS) for horizontal and vertical linear polarizations at 94 GHz. Unlike traditional single polarization measurements that retrieve only reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_w), this advanced radar also derives the linear depolarization ratio (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor. Table 1 provides a summary of the cited instruments, detailing their operational frequencies or wavelengths, the types of measurements they perform, their products, their measurement ranges, data collection intervals, and relevant references for a detailed description. The unique variables not processed by Cloudnet were humidity (q) and temperature (T) profiles, which were given by the microwave radiometer.

Table 1. Summary of instruments and their data products

Instrument	Bands / wavelengths	Measured	Products	Range	Time	References
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z , S_w	[13,51] m ^a	[1.1, 3.6] s ^b	Nils Kuchler et al.
MWR RPG	22-31 GHz (K-band)	Brightness	q and T	[10, 300] m ^c	[2, 2.5] min	Francisco Navas-Guzmán et al.
HATPRO	51-58 GHz (V-band)	Temperature	LWP	-	2 s	
CHM15k						
Nimbus Ceilometer	1064 nm		β	15 m	15 s	A. Cazorla et al.

Note that the range and time resolution of the cloud radar can vary in altitude and time. It depends on the radar chirp table configuration, where the time resolution, the maximum unambiguous velocity (Nyquist velocity), and the maximum range are defined. These three variables are interdependent and can be strategically adjusted to meet specific objectives. In our case, this table was modified several times throughout the analyzed period, changing these resolutions in a complex way along the entire time series. In addition, range resolution of temperature and humidity profiles given by the microwave radiometer also can be modified, therefore, it also vary with time.

1.1.2 Data processing

Instrument datasets are processed by Cloudnet processing chain (J. A. et al.), which sampled data on a common grid with a time resolution of 30 s and height bins based on the cloud radar range resolution. Taking into account the variability of the radar range resolution with time, to ensure consistency in the vertical pointing measurements, a regridding process was implemented to facilitate monthly statistical analysis when needed. The regridding involved interpolating measurements onto a height grid with

^aThis range vary with the height, and depends on the chirp table configuration
^bIt also depends on the chirp type
^cHumidity and temperature profiles has a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300m between 4.4 and 10 km

50 a 30 m resolution and was applied to every inter-annual profiling. Additionally, Cloudnet performs corrections for reflectivity gas and liquid attenuation, along with error estimation and quality flag assignment. Specifically, liquid attenuation corrections on radar reflectivity are crucial during raining events, as rain droplets can attenuate a significant fraction of the radar signal.

The processed database are then used as input to generate the target classification product, which will be used to classify clouds into different categories. This product assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed
55 grid. Each number are related to atmospheric elements in the following order: clear sky (0), droplets (1), drizzle or rain (2), drizzle & droplets (3), ice (4), ice & droplets (5), melting ice (6), melting & droplets (7), aerosol (8), insects (9), aerosol & insect (10). These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles is used to determine the 0°C isotherm and establish warm and cold regions to identify liquid and ice hydrometeors. Further, liquid droplets boundaries were resolved with lidar
60 attenuated backscatter coefficient (β), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, as the second are able to penetrate clouds and detect their tops. In addition, liquid layers respect a threshold of $T < -40$ °C, since it is not possible to keep water in liquid phase bellow that temperature. Moreover, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating
65 ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'Connor.

70 Despite the convenience of using Cloudnet products, target classification at the Granada showed instances of misclassification involving ice, melting, and rain pixels. Figure 2 illustrates a thinner and lower precipitable ice cloud present between 08:00 and 20:00. However, there is no evidence of cloud in the attenuated backscattering and radar reflectivity. These misclassifications stem from inaccurate representation of temperature profiles by models in this region, which subsequently affect cloud classification accuracy. To mitigate these issues and prevent misinterpretation of future cloud statistics, we refined the
75 classification algorithm (see section 2) to better handle these cases.

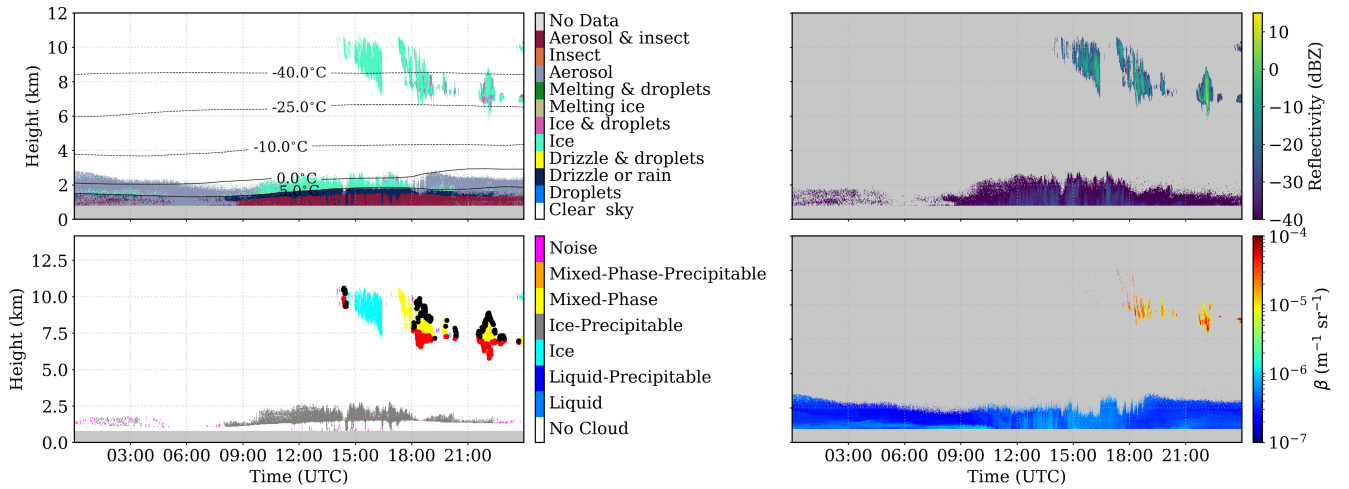


Figure 2. Temporary figure. A case study of Cloudnet hydrometeor misclassification. (a) Target classification product; (b) Radar reflectivity factor from Doppler cloud radar; (c) Attenuated backscattering coefficient from ceilometer. No evidences of clouds in the ABL can be identified by the instruments.

Also, specially important high level data such as liquid water content (LWC), ice water content (IWC), droplet effective radius and ice effective radius is described in table 2. For each product in this table, the necessary input data and methods used for their computation are specified alongside their associated uncertainty percentages and the studies from which these products are sourced. In general, spherical particles are assumed for both water droplets and ice particles. For droplets, a log-normal distribution is presumed, whereas empirical relations are utilized for ice particles.

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet products	Input role	σ_{re}/r_e	References
	Z and β : cloudb base and top identification		
Liquid water content (LWC)	T, P, assuming an adiabatic lifting MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T profiles	-35% to 90%	Robin J. Hogan et al. Julien Delanoë et al.
Droplet effective radius (r_{liq})	Z, and N and σ of lognormal PSD	15%	S. Frisch et al.
Ice effective radius (r_{ice})	Z and T	50%	Hannes Griesche et al.

The liquid water content (LWC) retrieval assumes a log-normal distribution for warm cloud droplets lifting adiabatically from its base. Its relation dependent on atmospheric temperature (T) and pressure (P). This method incurs a relative uncertainty of 15% as showed by A. S. Frisch et al.. At our station, T and P values utilized by the LWC retrieval are provided by the

European Centre for Medium-Range Weather Forecasts (ECMWF). The retrieval equation is applied exclusively to liquid pixels within cloud boundaries, which are determined by Doppler radar reflectivity (Z) and ceilometer-lidar attenuated backscatter coefficient. The liquid pixels, as ice, insect and aerosol pixels are given by the cloudnet target classification product, discussed in section ???. Additionally, LWC values are scaled to align with the Liquid Water Path (LWP) derived from the microwave radiometer. The liquid cloud droplet effective radius (r_{liq}) is retrieved by assuming a log-normal size distribution for warm cloud droplets. This method is insensitive to droplet number concentration and the logarithmic spread of the distribution (S. Frisch et al.). Consequently, its primary source of uncertainty is the reflectivity factor error, resulting in a relative error of 15% for r_{liq} retrievals. This relationship is applied exclusively to liquid pixels.

The ice water content (IWC) retrieval employs an empirical equation as determined by Robin J. Hogan et al., known as the Z-IWC-T relation, which depends on radar reflectivity and temperature. Compared to other microphysical variables, IWC retrievals has the largest relative uncertainties. For the W-band, the relative error is +55% to -35% for temperatures within the range of -20°C to -10°C, and +90% to -47% for $T < -40^\circ\text{C}$. While temperature is a significant parameter in IWC error estimation, Pablo Saavedra Garfias et al. demonstrated that the primary source of uncertainty is the reflectivity factor error. A typical reflectivity error of 3% can result in a 12% error in the ice water path (IWP), which is the integration of IWC over the atmospheric column. Therefore, errors in IWC can be relatively large. The Z-IWC-T relation is applied only to ice pixels within cloud boundaries. Next, the ice cloud particle effective radius (r_{ice}) is obtained following Julien Delanoë et al., who assumed the same empirical relation for IWC and the visible extinction coefficient (α) derived by Robin J. Hogan et al.. This approach, termed the Z-T relation, depends on both T and Z, and has a relative uncertainty of 50% of r_{ice} (Hannes Griesche et al.). In general, this high relative errors emphasize how challenge is retrieving ice properties accurately.

2 Methods

2.1 Cloud assessment by hydrometeor grouping

Many recent studies have been employed the algorithm of cloud classification by time (or by profile) with the aim of classify clouds (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.) and analyze their statistics. However, this algorithm admit different cloud typing inside the same cloud as a whole, assigning similar physical properties for different clouds. In the appendix ??, we describe this algorithm, adapting the criteria used by Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.. In another way, this section proposes a different classification algorithm, based on connecting groups of near hydrometeor.

First, the algorithm identify all hydrometeor pixels in the cloudnet target classification to generate a cloud mask. In sequence, hydrometeors with only two pixels of separation (at any direction) were assumed to be close enough to be considered part of the same cloud. Hence, a method of pixel dilatation and pixel contraction were applied to link these groups of closed hydrometeors. This method is explained in more details in appendix ??. Once the groups in target classification were identified, a sequence of verification is applied. First, the total number hydrometeor (without considering rain such as Drizzle or rain pixels) inside each group must be greater than 100 pixels to be considered as cloud. Otherwise, it was deemed as noise, being of unknown type.

For each cloud group, the percentage of each hydrometeor is calculated in order to apply the following classification scheme (shown in figure 3): if a cloud comprised more than 70% of Droplets and drizzle plus Droplets, it was classified as a liquid cloud (Liquid criteria). If it contained at least 10 connected rain pixels, it was further classified as a precipitating liquid cloud (Rainy criteria); otherwise, it remained classified as a non-precipitating liquid cloud. Similarly, if the previous verification was not satisfied, if this group contain more than 90% of ice or less than 10% of Droplets plus Ice & droplets, then it is classified as ice cloud (Ice criteria). Next, the same rain verification as in the liquid case is done to classify this group as purely ice cloud or precipitating ice clouds. Otherwise, cloud is classified as mixed-phase or precipitating mixed-phase through the same rain verification as other clouds. In relation of precipitating ice and precipitating mixed-phase clouds, an extensive analysis of individual case studies revealed that Drizzle & droplets only appears below the melting layer together with Drizzle & rain droplets. Therefore, to avoid an underestimation of cloud base height due to drizzle, we removed these pixels for these clouds since they are probably rain.

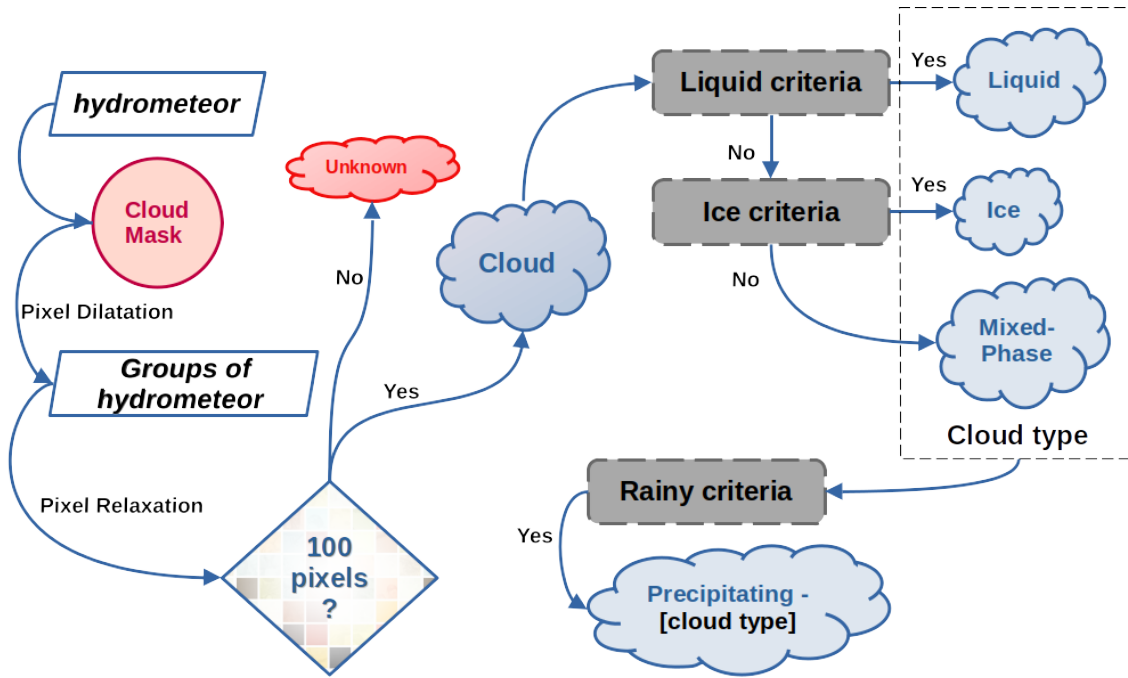


Figure 3. Scheme of the cloud classification algorithm by hydrometeor grouping

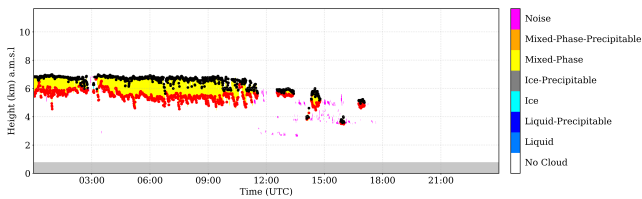
After the cloud classification, each cloud is then classified as single or multi-layer over time. Initially, time intervals where cloud overlap are identified. Overleaped pieces of clouds were classified as multi-layer, whereas non-overlapping pieces in principle can be classified as single layer. We then analysed cloud profiles for each cloud individually, acknowledging that a clouds can exhibit multiple layers with itself. If a cloud had a second layer separated by only one pixel at some time interval, that interval is identified as containing multi-layer. Otherwise, the clouds at that interval is classified as single layer. For the single layer case, cloud properties were calculated as follows: the cloud base and top height is determined by the first and last

height pixels within the cloud layer, respectively. Cloud thickness is obtained by the difference between cloud base and top
135 height pixels.

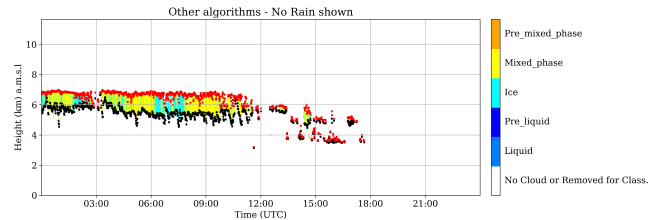
Taking into account ice, melting and rain hydrometeor misclassification around mid-day as discussed in section 1.1.2, which
can result into to low level ice and precipitating ice clouds misclassification, these cases were filtered. Basically, all these
clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700
meters were reclassified as noise. These values were determined empirically by studying many cases separately. Finally, for
140 those cases with no clouds, time interval is classified as clear sky condition.

2.2 Cloud assessment

A comparative analysis between the two classification algorithms were conducted in order to evaluated their applicability for
further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories
generated by these methods. The findings from this comparison will inform the selection of the most appropriate method for
145 future atmospheric studies and cloud analysis. In this framework, Figure 4 shows a case study at 6 May of 2023 in Granada
comparing cloud classification by time and by hydrometeor grouping for an mixed phase cloud, where red and black dots
represent the cloud base and top, respectively. As expected, the classification by time on the right panel lead to the fragmentation
of clouds into different types. In this case, the same cloud are being classified as ice and mixed-phase by time. On the other
hand, in the left panel, the classification by grouping preserves the homogeneity of clouds and classify it as mixed-phase only,
150 resulting in a more realistic classification.



(a) Cloud classification using the new method.



(b) Cloud classification using the old method.

Figure 4. Temporary figure (needs target classification). Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is time-based.

As a consequence, the time-based method of cloud classification can generate high correlations between cloud categories in
terms of frequency of occurrence and cloud properties. In relation to the cloud occurrence, right panel of Figure 4 suggest that
both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, this is occurring
because of the method of classification, which could affect further cloud statistic. To prove that, the 6-year dataset at the
155 Granada station was used to compute the daily average IWP for ice and mixed-phase clouds using the classification by time.
Figure 5 shows the closeness daily ice amount between these two clouds (mixed phase in above panel and ice cloud in the

bottom panel) and the similar pattern suggest a correlation generated by the cloud classification. It is worthy to note that this section is not focused on the analysis of IWP, this will be done in section XXX. Instead of this, here we centered on the correlation generated by the classification algorithms. Therefore, it was clearly observed that high correlation among cloud categories can generated similar physical properties among than, resulting in a misinterpretation of the results. It proceed because its difficult to attribute similar properties between clouds (formed by different mechanisms) as a realistic phenomenon that happens in the nature or attribute to the method of classification.

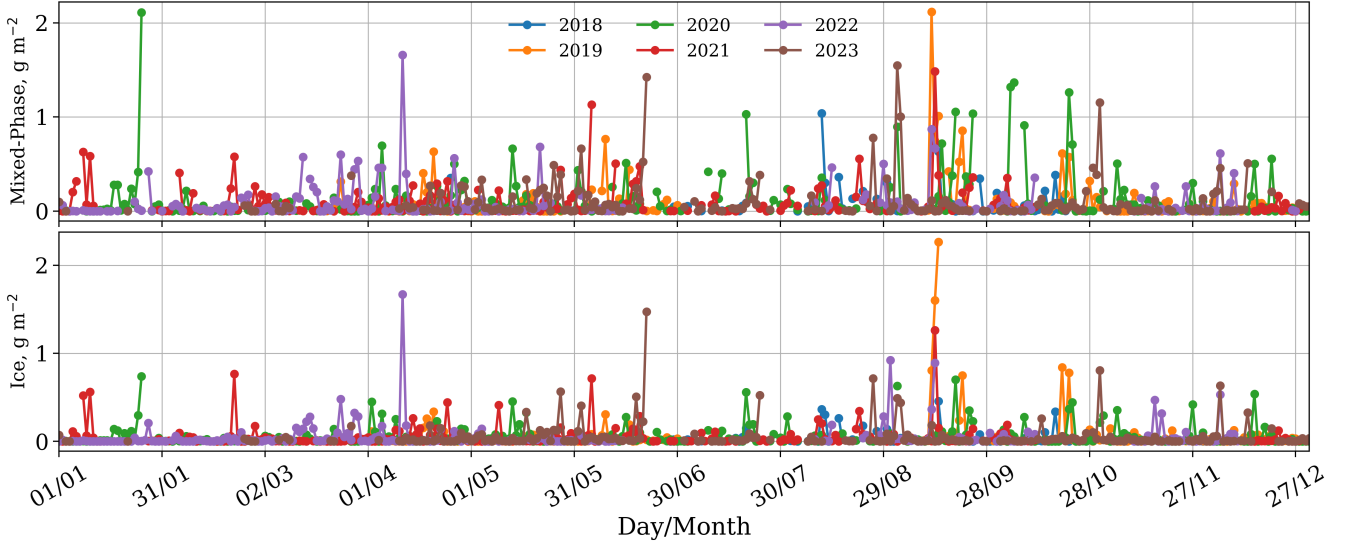


Figure 5. Time series of daily Mixed-Phase (top) and Ice (bottom) cloud ice water path from 2018 to 2023. Each year is represented by a different colored line.

Therefore, in order to quantify the correlation in between cloud categories generated by these two algorithms of classification, we calculate the daily cloud occurrence to access the correlation between cloud types. Figure 6 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the classification by hydrometeor grouping on the left, and for the classification by time on the right. As can be seen, the new proposed method demonstrate generally weaker correlations among cloud types, with most values clustering closer to zero. For example, the highest correlations observed were 0.19 between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, such as between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In addition, computed cloud properties will be not dependent on the method as they must be, avoiding some misinterpretation of the results.

In contrast, the classification by time exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed-Phase clouds was 0.54, and between Liquid and Mixed-Phase clouds was 0.44, highlighting significant interdependencies between these cloud types. Furthermore, the time-based method showed higher correlations between Liquid and

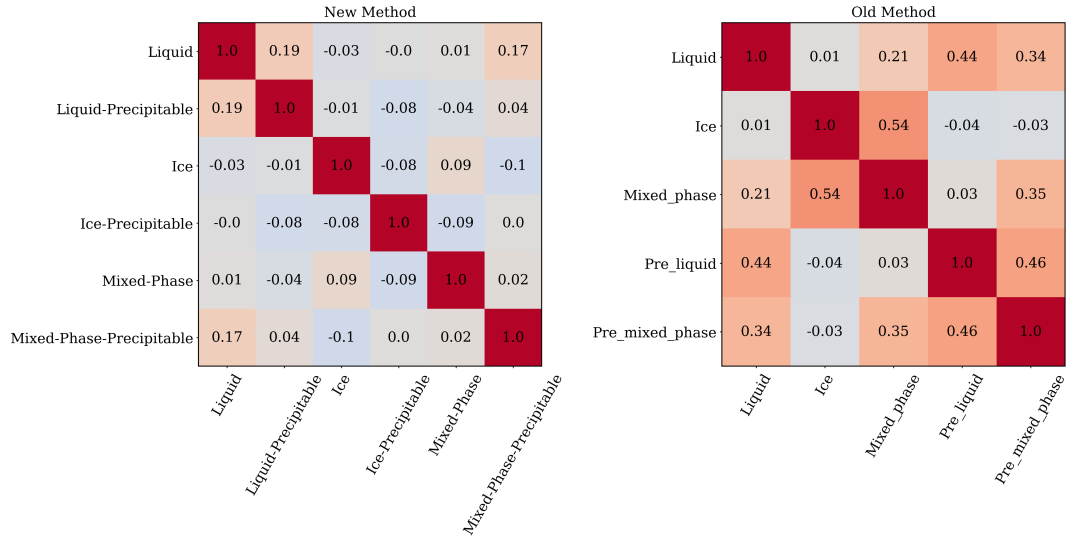


Figure 6. Correlation matrices comparing the daily occurrence between different cloud types using two methods. In the left, the New Method exhibits generally weaker correlations, indicating clearer distinctions between cloud types. In the right, The Old Method shows stronger correlations, suggesting more pronounced interdependence between cloud types

Liquid-Precipitable (0.34), Ice and Mixed-Phase-Precipitable (0.35), and Liquid-Precipitable and Mixed-Phase-Precipitable (0.46), indicating a less distinct separation between cloud types. These higher correlation values are a consequence of the classification scheme defined in this method. The cloud classification by time can fragment them, allowing tagging cloud profiles in different cloud categories, while they still inside the same cloud structure. Overall, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

3 Results and discussion

3.1 Annual variability and daily cycle of clouds

Since the cloud classification by hydrometers grouping showed a more comprehensive classification, this algorithm is employed in the following analysis. This section aims to analyze the inter-annual and daily cycle variability of cloud frequency of occurrence at the Granada station. Therefore, left panel of Figure 7 shows the monthly frequency of occurrence of different sky conditions, categorized into single-layer clouds, multi-layer clouds, clear sky, and noise. The data reveal a distinct seasonal pattern in cloud cover and clear sky occurrences. Clear sky conditions (green line) are predominant all year long, being most frequent in the summer months (80%), peaking in July and August, with a significant decline in spring (40%), reaching the lowest point in March and April. This pattern indicates a higher prevalence of clear skies during the warmer months

and increased cloudiness during the colder months. Single-layer clouds (blue line) exhibit a relatively weak seasonal pattern throughout the year, with a slight peak in spring (35%) and a decrease in the summer months (10%), suggesting a moderate variation in single-layer cloud occurrences. Multi-layer clouds (orange line) show a similar pattern, with a peak in March and April and a reduction during the summer, indicating a seasonal preference for multi-layer clouds in the spring. The noise category (red line) maintains a low and stable occurrence throughout the year, with minor fluctuations. Overall, the data highlight clear seasonal variations in sky conditions, with clear skies dominating in summer and increased cloudiness, both single and multi-layer, in the cooler months. Since this station exhibit a quit low cloud occurrence, a need to handle multilayer clouds is revealed to fill gaps of cloud properties at regions such as the western Mediterranean. Particularly, these multi-layer are problematic due to inaccurate LWP partitioning, leading a misclassification of liquid pixels inside clouds, as already discussed by Tatiana Nomokonova et al.. Therefore, the following analyses are focus only on single-layer clouds, seeing their physical properties can be retrieved more accurately.

Monthly frequency of occurrence is depicted in the right panel of Figure 7. Mixed-phase clouds (yellow) exhibit relatively consistent occurrences with slight fluctuations around 3% and 4%. Liquid-precipitable (dark blue) has a modest variation with zero occurrence in summer, followed by an increase during the transition to winter, reaching a maximum of 4% in November, suggesting increased precipitation during fall. Liquid clouds (blue) are more variable, showing a percentage of 3% in March and December, against zero in summer. Ice clouds (cyan) drops sharply in July and maintain almost constant levels around 5% and 7% throughout the rest of the year, indicating hotter temperatures that inhibit ice cloud formation during summer. Ice-precipitable clouds (gray) have a notable peaks of 8% in spring, followed by a decrease in summer (1%) and an increase in fall, until reach a local maximum of 5% during winter. It suggest a seasonal variations in ice-related precipitation with maximum in spring, unlike the case of water-related precipitation, which showed a maximum in November. Mixed-phase-precipitable clouds (orange) peak in April (6%) and December (7%), indicating higher chances of mixed-phase precipitation towards spring and the end of the year. Overall, ice, ice-precipitable, and mixed-phase clouds have the strongest seasonal pattern, indicating a higher sensibility of ice due to thermodynamic conditions than liquid droplets. Moreover, the precipitation in this region is intrinsic related to ice and mixed phase clouds i.e. with low temperatures.

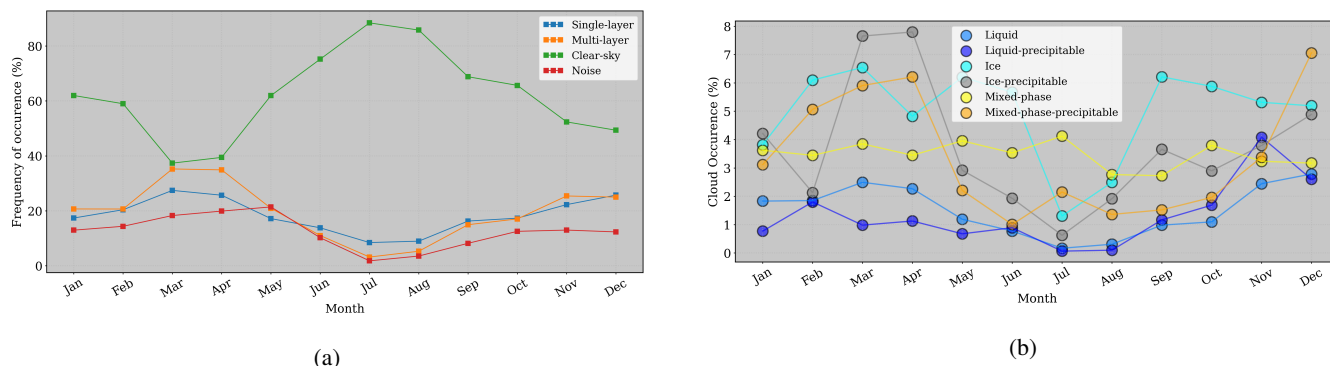


Figure 7. Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

An important mechanism associated with temperature decreases and responsible for extreme rainfall in Eastern Spain is the occurrence of cut-off lows (COLs). COLs account for nearly all rain events in Valencia, Spain (Ferreira), and are likely to affect the Granada station as well. This influence is attributed to the easterly wind component of COLs formed in the warm Mediterranean (Porcù et al.), which can reach the station, bringing humidity and cold air masses that favor cloud formation. In Valencia, Ferreira reported the rainiest days occurring in winter, with extreme rainfall events in spring and fall, aligning with our observations. During the summer, although almost non-precipitating clouds were observed at our station, temperatures were higher. Additionally, the Iberian Peninsula experiences a large low-pressure area over northern Africa, with high-pressure systems over the Atlantic Ocean and eastern and central Europe. This results in easterly winds over Andalusia, transporting moisture from the Mediterranean Sea (F. J. Bello-Millán et al.). This combination of high temperatures and humidity can create an environment conducive to storm formation. In this context, the few clouds observed in summer are likely related to convectively formed cumulonimbus clouds. This will be more evident when study their physical properties in further sections.

Next, the daily cycle of cloud occurrence should uncover the influence of temperature around mid-day, when the solar radiance is more intense. Although we already discuss the influence of synoptic and local phenomena in cloud formation, convection clouds should appear in the diurnal cycle. From Figure 8, its clear that there is no convection for any type of cloud during summer, fall and winter. Despite that fall shows an increase of ice clouds around mid-day, they are probably cirrus clouds, since their median base height and thickness are 8.5 km and 1.5 km, respectively (see Figure 9).

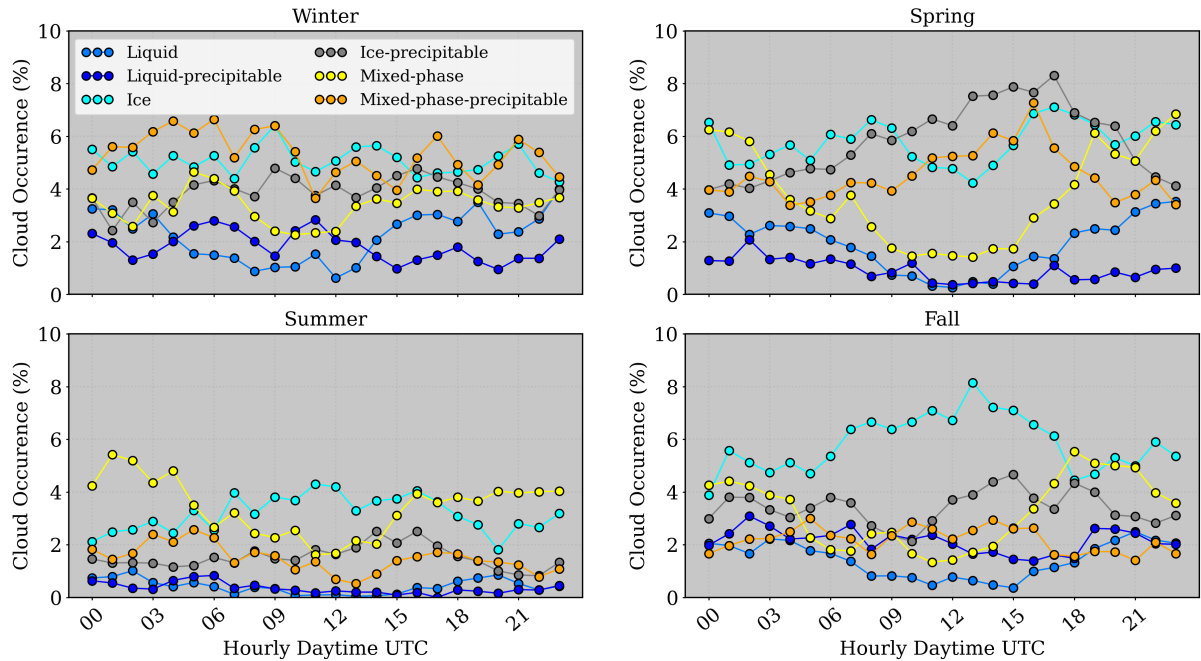


Figure 8. Inter-annual single layer cloud frequency of occurrence. The frequency of each cloud type is indicated in the legend.

However, spring shows a possible presence of convective clouds, with a slight peak of occurrence for Ice-precipitable and Mixed-phase-precipitable clouds between 09:00 and 21:00 UTC. Different from non-precipitating clouds which reached a minimum around mid-day, these clouds doubles the occurrence from 4% at 00:00 to 8% at 16:00-17:00 UTC. These observations
235 are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

240 3.2 Seasonal patterns in cloud base, top, and thickness

Seasonal evolution of cloud base height, top height and thickness heights are displayed in Figure 9. This figure shows that precipitating clouds exhibit a significantly lower cloud base height (top panel) compared to non-precipitating clouds, as well as a narrower distribution. Regarding non-precipitating ice clouds, they possess the highest cloud base heights, with median values around 8 km. These high ice clouds with small ice water path (Figure 22) can be associated to cirrus clouds. Next,
245 non-precipitating mixed-phase clouds are the second highest cloud type we found, which show considerable variability in height except during summer, when values stabilize at 6 km. Differently, cloud base is high in fall, where the 50th percentile reach a maximum of 7 km of height, along with high variability. On the other hand, non-precipitating liquid clouds showed higher cloud base variability during summer, achieving significantly different heights compared with other seasons. Overall, in summer, liquid clouds, precipitating mixed-phase clouds, and ice precipitating clouds demonstrate elevated cloud base heights,
250 attributed to the relatively high temperatures and low humidity at this season. Specifically, liquid clouds display the highest cloud base variability during summer, with the 50th and 75th percentiles around 4 km and 6 km, respectively. These clouds can be related to stratocumulus and stratus clouds, which were dis-coupled from cumulonimbus at final stage.

As observed for the cloud base, precipitating clouds have a significantly lower median cloud top height (middle panel of Figure 9) compared to non-precipitating clouds and exhibit a narrower cloud top distribution, except in the case of ice
255 clouds. Non-precipitating ice clouds maintain a similar distribution across seasons, highlighting another feature that suggest the presence of cirrus clouds, while ice-precipitable clouds demonstrate highly variable cloud tops. Given their well-defined cloud bases, these clouds exhibit greater cloud thickness, indicative of storm activity at the station. The cloud tops for liquid clouds mirror the variability of their cloud bases, particularly during summer. These clouds exhibit a well-defined cloud thickness, with purely liquid clouds having a thickness below 1 km and liquid-precipitable clouds reaching up to 1 km in fall, as indicated
260 by the 75th percentile. This aligns with the observed peak occurrence of these clouds in this season. Mixed-phase clouds also follow the behavior of their cloud bases and display a well-defined cloud thickness. Generally, non-precipitating mixed-phase clouds have a thickness around 500 m, increasing to 1 km for precipitating mixed-phase clouds, with slightly higher values in summer. For non-precipitating mixed-phase clouds, median cloud tops vary around 7 km during spring, summer, and winter, but reach 8.5 km in fall. Precipitating mixed-phase clouds have cloud tops around 3 km in spring, fall, and winter, peaking at
265 7 km in summer.

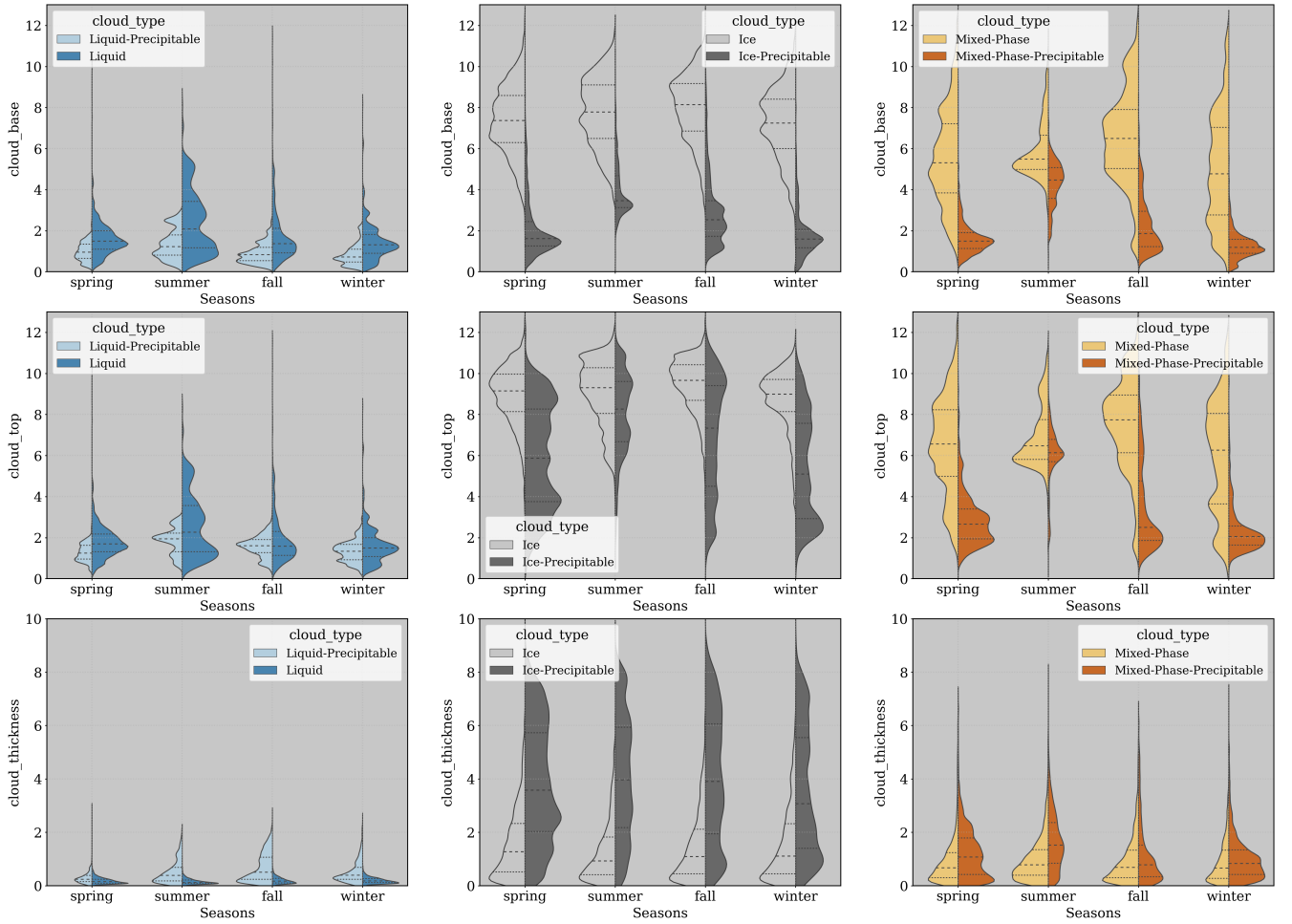


Figure 9. Seasonal evaluation of cloud thickness, base and top. The legend above indicates the color correspondent to each cloud type. This figure shows a dark gray box-plot with a white median inside an estimated density distribution. This distribution is displayed in relation to the middle of the box-plot in the right and the left direction.

Overall, ice-precipitable clouds exhibit the greatest cloud thickness, followed by ice, mixed-phase-precipitable, mixed-phase, liquid-precipitable, and liquid clouds. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops might be underestimated due to high attenuation in some cases, reducing radar sensitivity.

3.3 Cloud microphysics properties

270 3.3.1 Liquid clouds

Liquid clouds shows high variability of LWC in function of the height. The domain with elevated LWC is marked by hot colors and white lines, which marks constant values of 1000 mg m^{-3} (Figure 10). These high values represent just a few

observations, and are hidden in seasonal median profiles above 2 km on the left panel. Only values at low levels during spring were clearly seen by pronounced peaks surpassing 250 mg m^{-3} . Overall, during colder months (winter and spring), LWC for liquid clouds can vary from 50 to 1000 mg m^{-3} . On the other hand, LWC in summer is sparse but can be found at higher levels as expected. Warm months at this region has high temperature and less relative humidity (?). Next, average liquid water path (LWP) (bottom panel of Figure 10) demonstrates that from January to June, and also early fall, there is a considerable variability in LWP, highlighting how different liquid clouds can be in these months. It is clearly observed by differences between the average (marked as red line), and the median (marked as the black star). A notable increase in the median LWP occurs in April, aligning with their maximum cloud occurrence. From August to October, there is a gradual increase in LWP, with a noticeable difference between the median and the mean. This trend corresponds with the increased occurrence of liquid clouds in fall, which is impacted by different mechanisms. Since we reported a few occurrences of these clouds, their statistics will be much more sensible to these mechanisms.

Seasonal median profiles of LWC show high variability with height and a pronounced peak around 300 meters in spring. This variability reveals that there is no significant difference of the amount of water in liquid clouds between seasons. In general, the magnitude of the LWC is relatively small, and values are fluctuating around 70 mg m^{-3} in the entire vertical, except about 300 meters. It is not shown here, but at this level, the median values are strongly influenced by liquid clouds with high amounts of water in spring 2020 ?. Undoubtedly, the modest frequency of occurrence of this cloud category makes their microphysics quite variable as already observed. Next, although LWP in summer are the smallest, there are some peaks identified at 2 and 5 km. This season presented no considerable difference between the average and median LWP (bottom panel) except for June, showing that these values are representing just a few occurrences of this cloud, which can be checked by the blue line. This line shows the number of cloud profiles used to calculate the statistics per months.

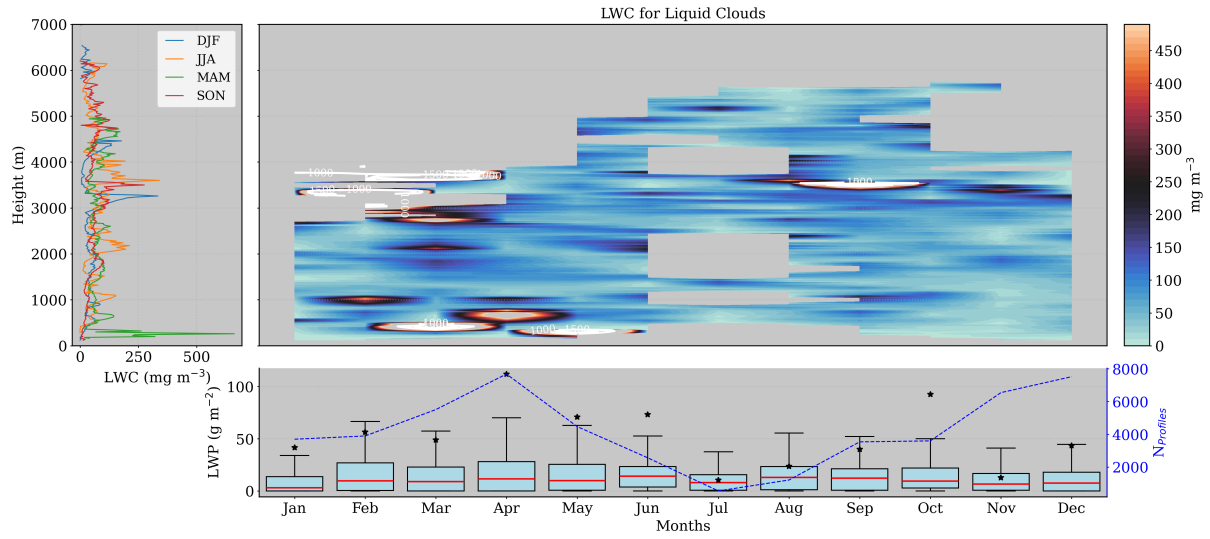


Figure 10. Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal mean profiles of LWC. Right panel shows the monthly mean profiles of LWC and monthly box-plot of LWP. For the boxplots, medians are denoted by red lines and the average by black stars.

In terms of droplet effective radius, liquid clouds are well consistent throughout the year, showing small differences in terms of median values (top panel of Figure 11). There is no discernible pattern in r_{liq} across seasons, with higher altitudes weakly correlating to larger droplet sizes. Generally, measurements above 4 km are sparse, occurring primarily in summer and early fall. Boxplot analysis indicates a stable median r_{liq} of approximately $5 \mu\text{m}$, with a small decrease in July, when there is practically zero occurrence of these clouds.

Seasonal profiles confirm the uniformity in r_{liq} distribution, showing minimal variation with altitude. The values are slight higher than $5 \mu\text{m}$ across all seasons, suggesting consistent cloud composition and droplet size throughout different atmospheric conditions. These findings underscore the stable nature of liquid clouds in terms of droplet size and distribution.

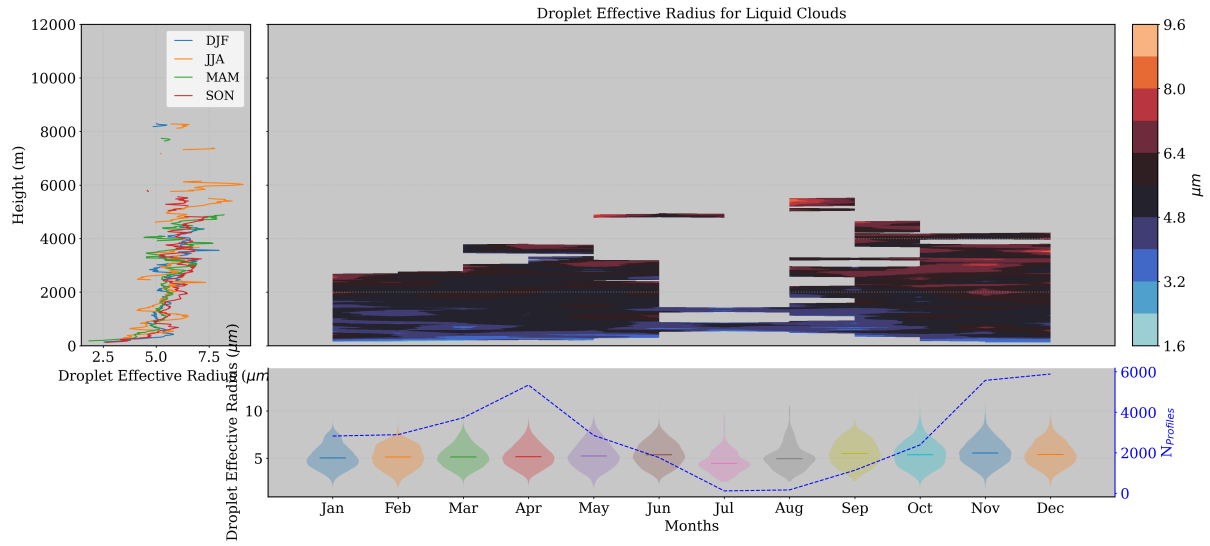


Figure 11. Seasonal and monthly statistics on droplet effective radius (r_e) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius, where shaded area denote the 25th and 75th percentile. Top-right panel shows the monthly profiles of the median r_e , while the bottom-panel displays the size distribution of r_e by months.

Afterwards, significant liquid water content (LWC) for precipitating liquid clouds is primarily found below 3 km (top-right panel of Figure 12), consistent with their median top heights. Regions with significant values of liquid water content show broader areas, which is expected as these clouds are thicker than purely liquid clouds. Monthly trends indicate that higher amounts of liquid water content occurs in later fall, winter and spring. However, the highest amount of LWC were identified in
305 June (early summer), very close to the ground (300 m). Then, in the rest of summer and early fall, atmospheric water content is almost zero. In spring, significant values is found around 500-1000 m, while in winter, they are above 1 km. At these regions, LWC exhibited a strong variability, ranging from 200 and 500 mg m^{-3}

The bottom panel of Figure 12 below indicates substantial variability of LWP in January, suggesting diverse variety of liquid clouds during this month. In contrast, spring exhibits relatively consistent LWP across its months, reflecting a stable period for
310 precipitating liquid clouds. In these months, their median stays below 50 g m^{-2} , and summer have even less LWP, except for June, as previous observed.

Seasonal median profiles of LWC emphasizes a clear increase of LWC in function of height during fall, winter and summer. For the last, two peaks of 250 mg m^{-3} are clearly identified at 2 and 3 km, respectively. This season presented more variability with height, showing more peaks of LWC in function of height. Falls also presents a peak of 250 mg m^{-3} at 2.5 km, and winter
315 presents the peak at 2 km.

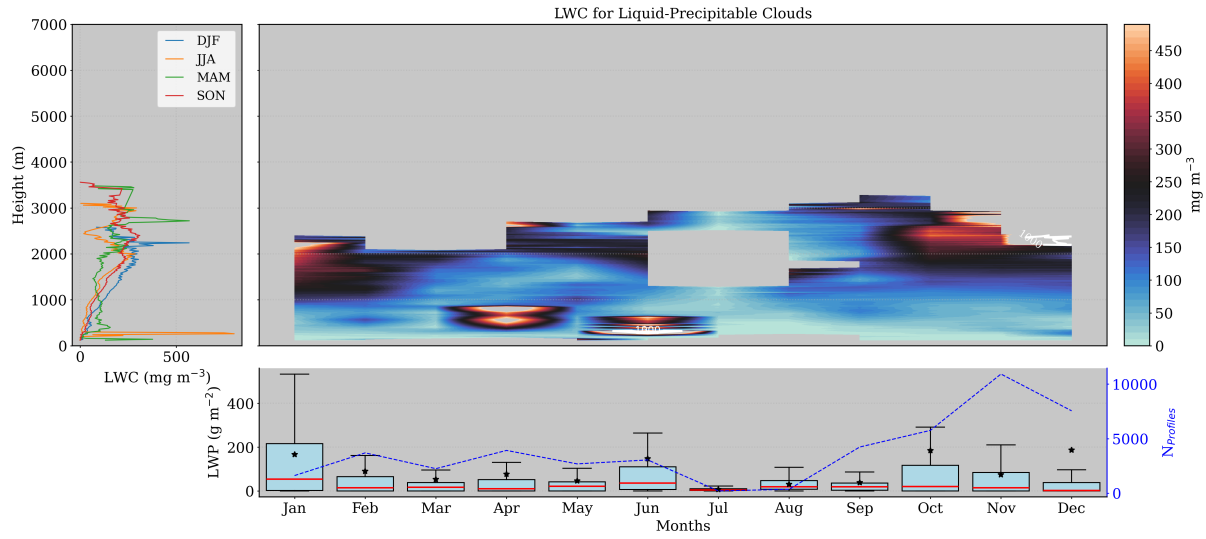


Figure 12. Same as Figure 10, but for precipitable liquid clouds.

Different for the purely liquid clouds, precipitating liquid clouds demonstrated radically different seasonal patterns (see Figure 13). In spring and summer, their r_{liq} are effectively small then fall and winter, as can be seen by light and hot colors. The r_{liq} in fall and winter often reaches $15 \mu\text{m}$ below 2 km. Despite the fact reflectivity was corrected by liquid attenuation by Cloudnet, challenges persist, particularly in intense rain events where accurate r_{liq} determination is compromised, possibly influencing wet radar radomes.

Their monthly size distribution in bottom panel displays different size distributions over the year, where higher median values are found in fall as already observed, specially in November. During spring and early summer, the median is maintained around $10 \mu\text{m}$, decrease in summer, and increase again in fall.

Seasonal profiles further elucidate r_{liq} dynamics. Summer and winter exhibit similar r_{liq} values up to 2 km altitude, around $10 \mu\text{m}$, decreasing to approximately $5 \mu\text{m}$ at 4 km. Spring shows a distinctive pattern with r_{liq} increasing from $5 \mu\text{m}$ to a peak of $15 \mu\text{m}$ around 3 km altitude, suggesting seasonal variability in cloud composition and droplet size distribution.

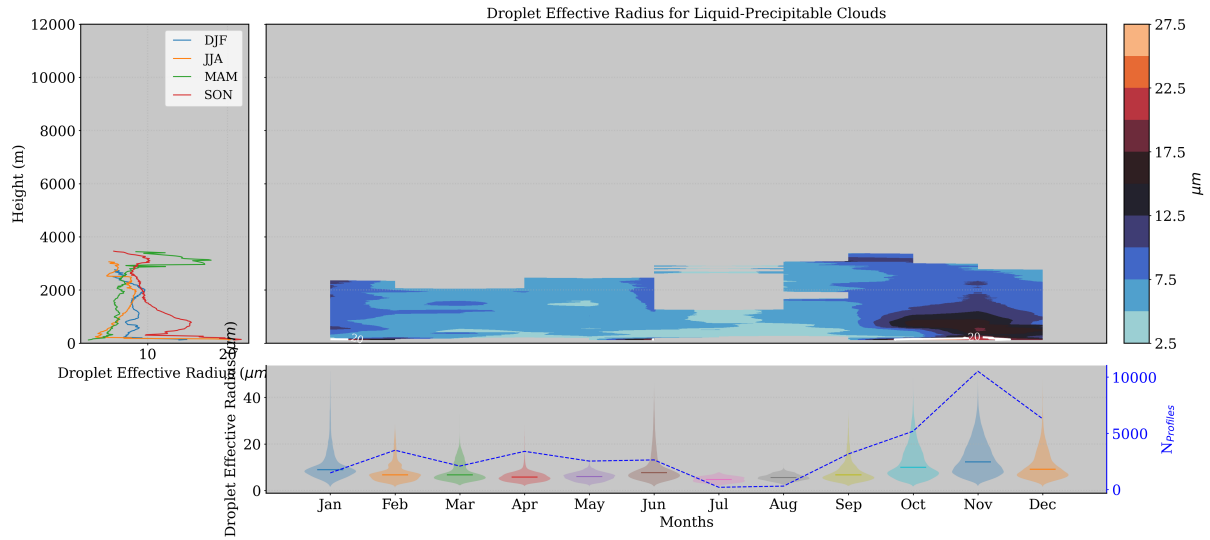


Figure 13. Same as figure 11, but for precipitable liquid clouds.

In general, precipitating liquid clouds has more liquid water content than only liquid clouds and are centered at lower levels. Liquid clouds presented high variability with height and months with more elevated water content. However, these peaks are representing just some case at isolated levels, and the seasonal median profiles do not turn in evidence since others months have much less water at the same levels. On the other hand, precipitating liquid clouds in general has more liquid water content over the months. As a consequence, integrated liquid water content is greater for precipitating liquid clouds. In addition, the droplet effective radius are greater for precipitating liquid clouds since they have more drizzle particles.

3.3.2 Mixed Phase clouds

Next, mixed-phase clouds indicates generally lower liquid water content (LWC) values compared to liquid-phase clouds (top-right panel of Figure 14). These clouds have a consistent occurrence throughout the year (around 3-4%), but exhibit very high variability of LWC in function of height, as expected since their cloud base also has such variability. Specially in summer and winter, high LWC are encountered above 8 km, surpassing 1000 mg m^{-3} . As for the liquid cloud case, these values occurs just few times at this level, does not affecting the seasonal median (left panel of Figure 14), since other summer-months has much smaller LWC at this height. In addition, this month was registered as having presented cumulonimbus clouds (XXX cite something) events, suggesting that liquid phase water can be related with these clouds.

Following, although LWC varies greatly during winter, spring and fall, with most pronounced occurrence below 4 km, their integral remains really close to zero. This can be observed by the bottom panel of Figure 14, which shows an absence of LWP throughout the entire year in terms of the median. On the other hand, just a few cases with elevated values are generating high mean LWP, highlighting the presence of outliers. Next, vertical profiles of median LWC by season denoted high variability below 4 km and above 10 km, for all stations. Between this level, LWC is relatively small at all seasons, fluctuating around

zero. Otherwise, median LWC have local maximums around 100 mg m^{-3} , with two notable peaks surpassing 500 mg m^{-3} at 500 m during spring and summer (XXX explain what happened). These seasonal variations underscore the complex and dynamic nature of mixed-phase cloud formation and their differing behaviors across various atmospheric conditions.

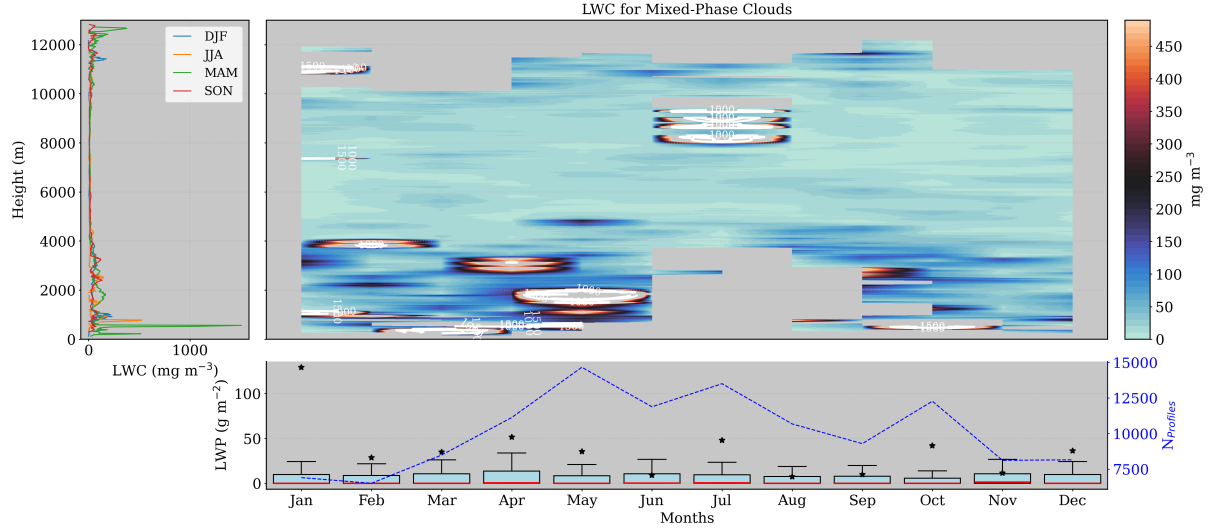


Figure 14. Same as Figure 10, but for mixed-phase clouds.

In terms of droplet effective radius (r_{liq}), a totally different nature is observed in relation to LWC. The last has expressive amounts of LWC at low and high layers ($< 4 \text{ km}$ and $> 10 \text{ km}$), while the first is exhibiting high droplet sizes at middle levels. Large values of droplet sizes can be observed on top panel of Figure 15 in dark blues and hot tones. They are normally located between 4 and 8 km throughout the entire year, except for July, which also shows larger droplets at 10 km. As already discussed, liquid water are not common at those levels, but when detected, are followed by high mass concentration, suggesting the detection of cumulus clouds. Overall, summer presents higher droplets compared to colder months, with effective radius between $14\text{--}22 \text{ }\mu\text{m}$ at medium heights ($> 4 \text{ km}$ and $< 6 \text{ km}$), and between $12\text{--}18 \text{ }\mu\text{m}$ around 10 km. On the other hand, droplet size are generally close to $10 \text{ }\mu\text{m}$ inside these clouds. This is shown by the monthly distribution of r_{liq} , where the mode are located at even small values. In fact, cloud droplets are slight larger in summer and has good probability to reach $15 r_{liq}$ over the year, but in general they have more small droplets.

In addition, there is a clear dependency of droplets size with the height. Seasonal profiles on the left panel of Figure 15 shows droplet size increasing up to 5 km, then starting to decrease. Thus, it is followed by all seasons, indicating that particular seasonal differences of droplet size is only grasp by monthly profile analysis. However, the seasonal profiles highlights a general behavior, which can be explained by the evaporation of liquid cloud droplets trough air particles injection into clouds. The injection of particles will increase the water vapor competition, evaporating some droplets while ice particles are also formed due to low temperatures. This factors will decrease the size of water droplets and increase the mass concentration of ice particles. The last can be confirmed by left panel of Figure 17.

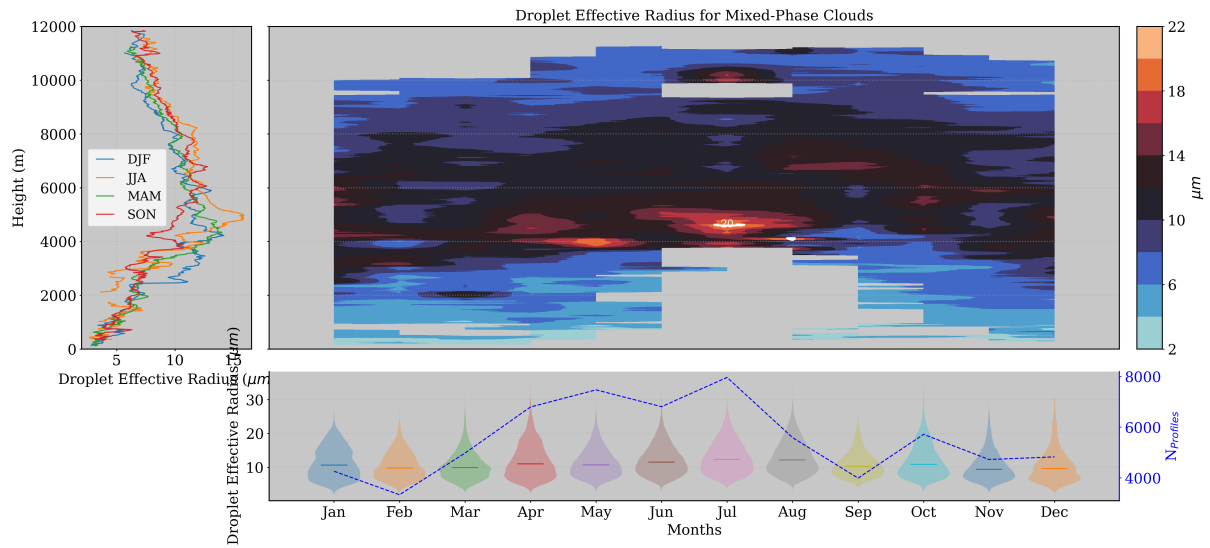


Figure 15. Same as figure 11, but for mixed-phase clouds.

Regarding the ice water content, it exhibit significant variation with height, especially during summer. The monthly median profiles in Figure 17 shows that ice mass is greater at middle/high altitudes, as expected. Normally, significant values of IWC of about $20\text{--}40\text{ mg m}^{-3}$ are found between 4 and 10 km, except for July. As for the LWC, July also presented elevated IWC above 10 km (more than 60 mg m^{-3}), complementing our hypotheses that this clouds can be classified as cumulus clouds.

370 Cumulus clouds are the first stage of cumulonimbus clouds and can have considerable amounts of ice, specially at their top. In addition, these clouds have a considerable thickness as those observed the for mixed phase clouds (some of them can reach 2 km of thickness). In general, these clouds have low ice amounts as can be seen by the median in the bottom panel, normally below 10 g m^{-2} throughout the entire year. However, the presence of just a few cases with high amounts of ice (outliers) can pull the average up. It happens specially in April and summer, where the physical properties of these clouds are more variable.

375 As already observed, IWC increase with the altitude. It is generally formed above 2 km, and then its mass concentration started to increase until 7 km for colder months (winter and spring) and 8 km for warmer months (fall and summer). Winter and fall follows practically the same function of height, as well summer follows the same behavior of fall up to almost 8 km. Above this level, summer has the largest amount of IWC, with maximun, varying between 20 and 10 mg m^{-3} .

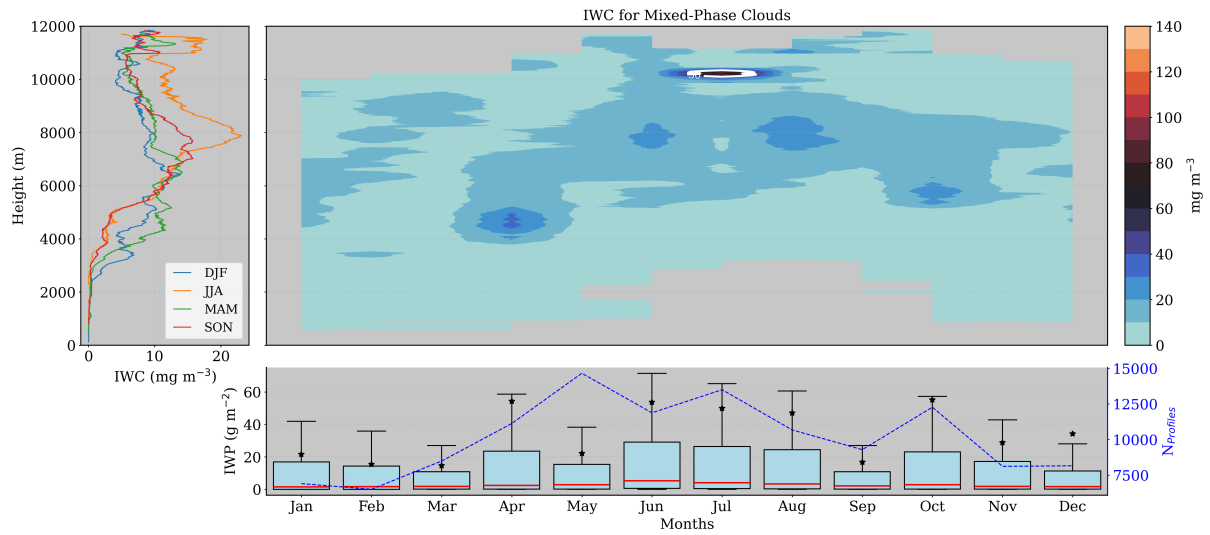


Figure 16. Same as Figure 22, but for mixed-phase clouds.

In terms of effective radius, their size follows almost the inverse of the IWC in function of the height. Figure 17 undoubtedly shows that higher radius are at low levels, except in July where its possible to find (r_{ice}) larger than $50 \mu\text{m}$ above 10 km. Hence, the general properties of these clouds can be described as having larger ice particles with low ice water content (IWC), particularly at lower altitudes. This characteristic is not limited to summer, during which these clouds are distinctly different from those observed throughout the rest of the year. In addition, the monthly distribution (bottom panel of Figure 17) shows that larger ice effective radius are more frequent than small particles. The median are normally fluctuating around $45 \mu\text{m}$, with the mode above this value.

The seasonal profiles on the left panel also strengthened our finding that effective radius decrease with height. Actually, it still almost constant with height up to 6 km, where it starts to decrease. It values in summer are slight higher but there is no significant differences over the seasons. Normally, the r_{ice} is around $30 \mu\text{m}$ above 10 km, except for July.

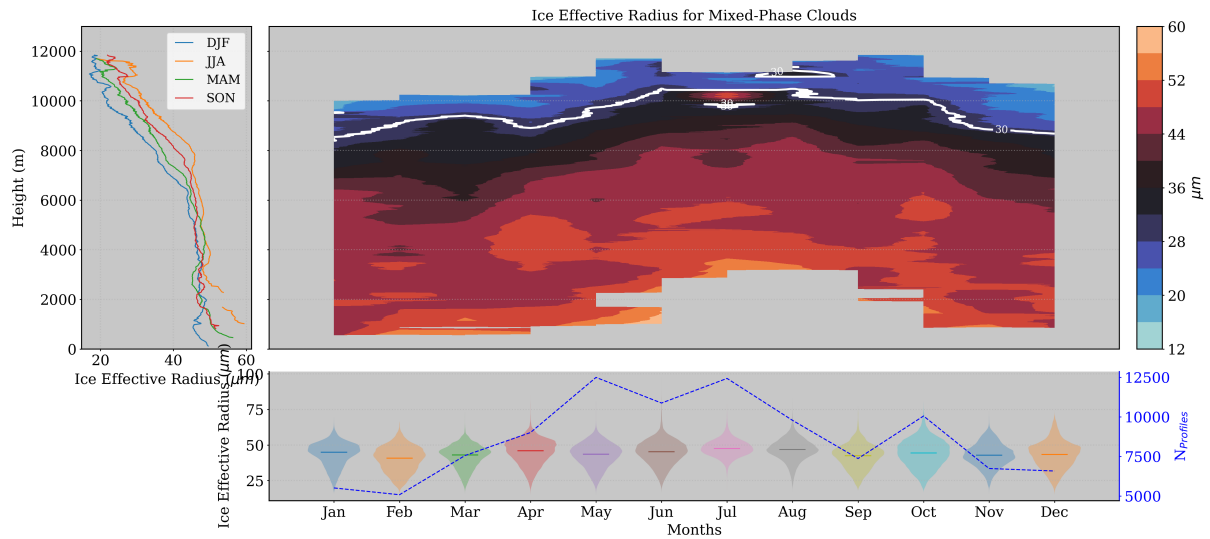


Figure 17. Same as figure 11, but for mixed-phase clouds.

The contour plot of liquid water content (LWC) indicates significantly higher values compared to non-precipitating mixed-phase clouds, attributable to the elevated presence of supercooled water droplets. Local maxima align with the months of highest cloud occurrence, specifically late fall, spring, and winter. During these months, high LWC values are concentrated below 2 km, though some values around 100 mg m^{-3} are observed near 4 km. Within the sub-2 km range, LWC varies from 80 to 400 mg m^{-3} , with peaks identified in October, December, January, and April. In contrast, summer shows no significant water amounts compared to other months. Next, liquid water path (LWP) reveals even greater discrepancies between the mean and the median compared to non-precipitating mixed-phase clouds. The mean LWP is influenced by a few instances of clouds with high LWC, while the median remains constant and close to zero throughout the year. Fall, winter, and spring display mean values ranging from 100 to 200 mg m^{-2} , with median values around 0 mg m^{-2} . Summer exhibits less variability, reflecting the infrequent occurrence and generally low LWC of these clouds during these months.

Seasonal profiles of LWC further underscore these patterns. Summer records 0 mg m^{-2} of LWC across different heights. For other seasons, LWC values are confined below 2 km, with a peak around 1 km. Fall and spring show slightly lower mean LWC values close to 200 mg m^{-3} , while winter reaches higher values, around 250 mg m^{-3} . These observations highlight the seasonal variability in liquid cloud characteristics and their dependence on atmospheric conditions.

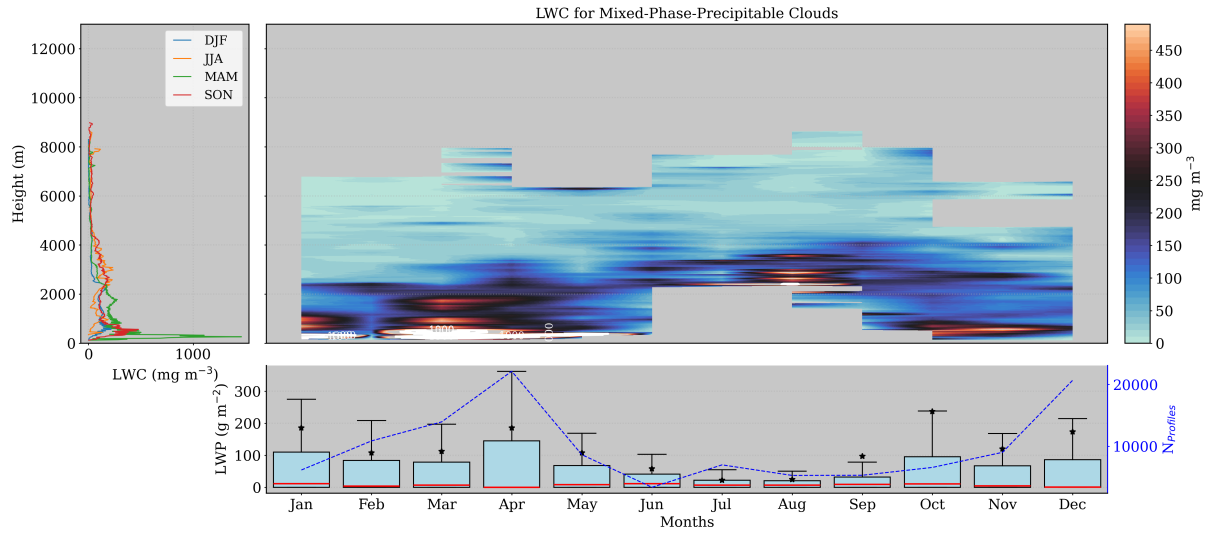


Figure 18. Same as Figure 10, but for precipitable mixed-phase clouds.

Analysis of liquid water content in clouds with precipitation characteristics reveals distinct features compared to non-precipitating clouds. These clouds exhibit higher ice water content (IWC) and significant variability in both altitude and temporal distribution. Local maxima in IWC were observed during winter, April, and summer, predominantly between 3 and 8 km altitude. A notable peak of about 200 mg m^{-3} was observed in June at 9 km, surpassing similar peaks in the non-precipitating case, indicating the presence of recently formed convective clouds that transition into non-precipitating clouds in subsequent months.

Boxplot analysis of liquid water path (LWP) indicates that while monthly median values are typically near zero, the low occurrence frequency in summer and early fall results in higher comparable values compared to other seasons, which generally have more clouds with smaller amounts of ice.

Seasonal profiles further delineate the characteristics of these clouds. Winter and spring exhibit similar IWC peaks around 4 km, with values approximately 120 mg m^{-3} , reflecting consistent conditions conducive to precipitation. Summer shows slightly higher maximum IWC compared to fall, reaching peaks at 6 km and 5 km, respectively, with values close to 200 mg m^{-3} . Additionally, summer features a pronounced secondary peak around 9 km with IWC peaking at 150 mg m^{-3} , highlighting the presence of deep convective processes and indicating complex cloud dynamics during this season.

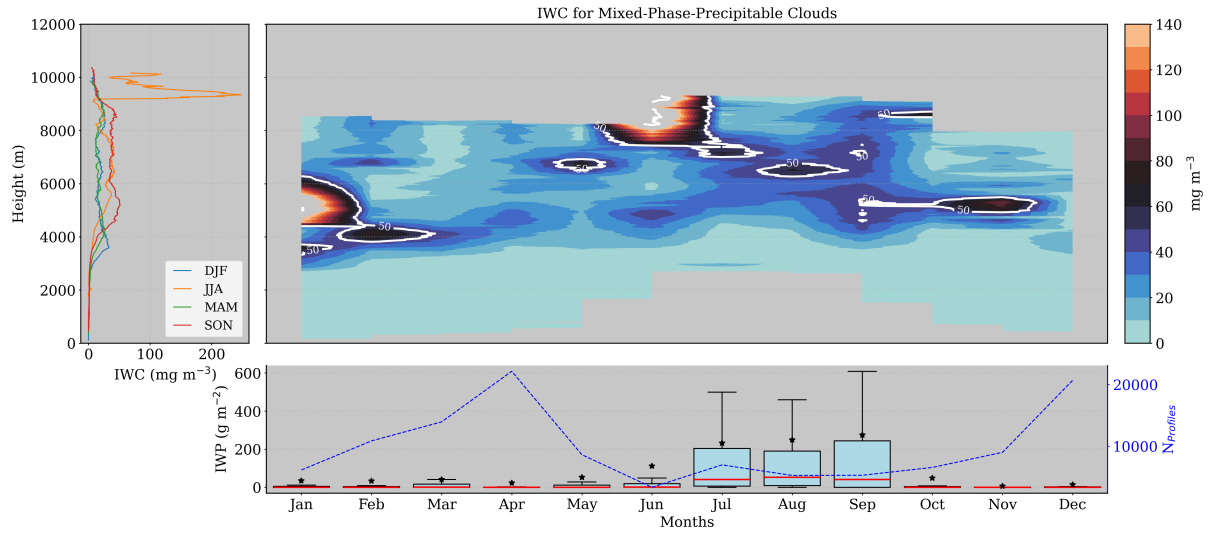


Figure 19. Same as Figure 22, but for precipitable mixed-phase clouds.

The analysis of droplet effective radius (r_{liq}) in cumulonimbus clouds reveals intriguing seasonal and altitude-dependent dynamics. Throughout the year, the highest r_{liq} values are consistently observed near the surface, indicative of localized processes such as drizzle particle formation. Summer stands out with notably elevated r_{liq} values ranging up to $20 \mu\text{m}$ from ground level to 3 km altitude, attributed to the presence of large droplets typical in cumulonimbus clouds. These findings underscore the impact of strong liquid attenuation on r_{liq} measurements, complicating accurate retrieval methods. Boxplot analysis highlights stable median r_{liq} values around $10 \mu\text{m}$, yet exhibits significant monthly variability, peaking above $20 \mu\text{m}$ in winter and spring, and surpassing $30 \mu\text{m}$ in summer. Seasonal profiles further illustrate this variability, with r_{liq} values decreasing up to 2 km altitude, briefly increasing around 4 km, and then decreasing again, emphasizing the complex microphysical processes within cumulonimbus clouds. These insights emphasize the need for refined observational techniques to accurately capture and interpret r_{liq} in varying meteorological conditions.

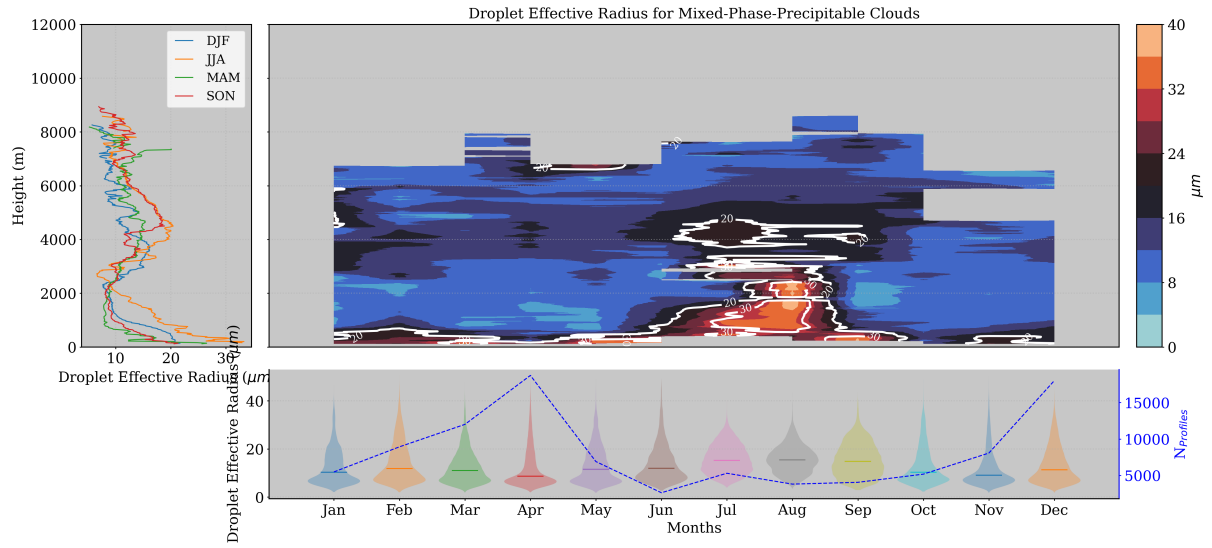


Figure 20. Same as figure 11, but for precipitable mixed-phase clouds.

The study of droplet effective radius (r_{liq}) in various cloud types reveals significant insights into seasonal and altitude-dependent variations. Cumulonimbus clouds consistently exhibit higher r_{liq} values near the surface, influenced by local processes like drizzle particle formation. Summer months notably feature larger droplets up to $20 \mu\text{m}$ from ground level to 3 km altitude, indicative of intense convective activity. Despite stable median values around $10 \mu\text{m}$, monthly distributions highlight considerable variability, peaking in winter and spring and reaching extremes exceeding $30 \mu\text{m}$ in summer. Seasonal profiles underscore these dynamics, with r_{liq} decreasing up to 2 km, briefly increasing near 4 km, and then declining, reflecting complex microphysical interactions within cumulonimbus clouds. These findings emphasize the necessity for precise measurement techniques to accurately capture and interpret r_{liq} across diverse meteorological conditions.

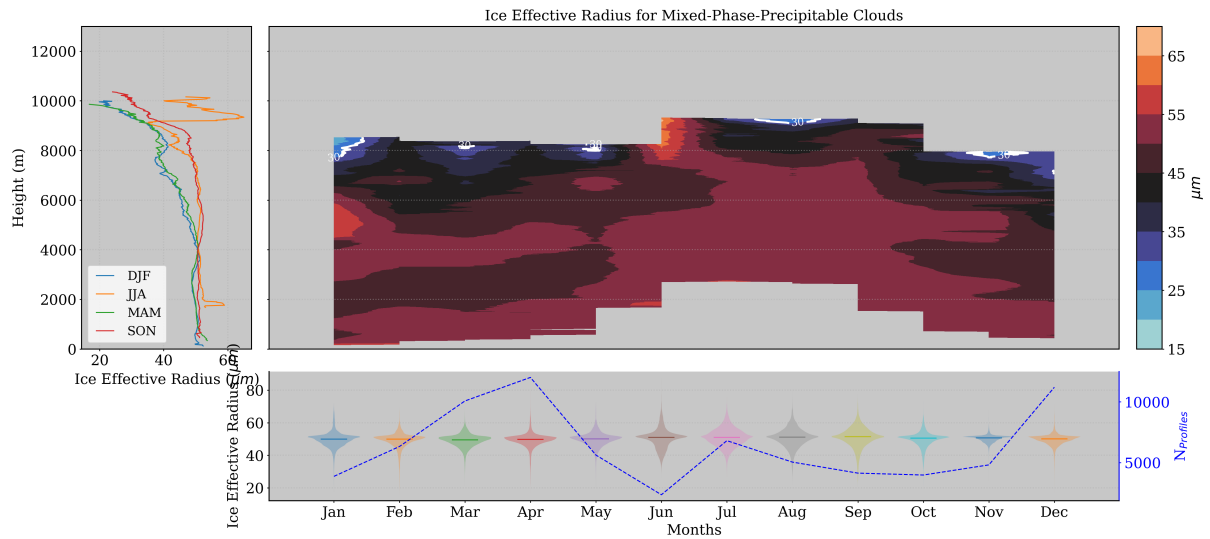


Figure 21. Same as figure 11, but for mixed-phase clouds.

3.3.3 Ice Clouds

Our analysis of ice cloud liquid water content (IWC) reveals distinct seasonal and altitudinal variations. The majority of ice clouds have bases around 6 to 8 km; however, the contour plots indicate significant IWC values predominantly between 4 and 8 km, except for January when extremely high values were detected as low as 2 km. This anomaly in January is attributed to two extreme weather events: the Filomena storm in 2020 and the Gloria storm in 2021, which significantly impacted the western Mediterranean region Marta de Alfonso et al.; Enrique Pravia-Sarabia et al.; Marcos García Sotillo et al.. Ice clouds at altitudes of 4 to 8 km typically exhibit IWC ranging from 100 to 300 mg m^{-3} during winter, the transition to summer, and fall. Boxplot analyses of the liquid water path (LWP) show the highest variability in January due to these extreme snow events, which skew the statistics, usually characterized by low median IWC values. Mean IWC values generally hover around 50 mg m^{-3} in spring, fall, and winter, influenced heavily by the Filomena and Gloria storms, while late summer values approach zero, with almost no ice clouds observed during this season. Seasonal profiles further support these findings, showing consistent IWC values between 4 and 8 km during spring, early summer, and fall, with peaks around 200 mg m^{-3} . Notably, early winter profiles reveal a significant peak exceeding 500 mg m^{-3} at 2 km, corroborating the impact of the extreme snow events observed in January

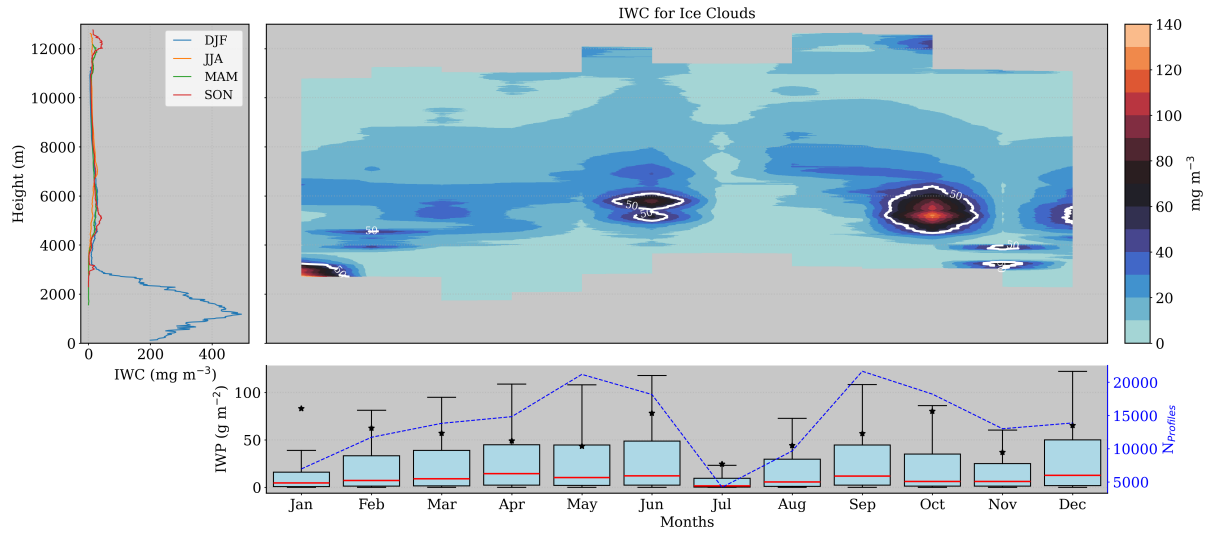


Figure 22. Seasonal and monthly average of ice water content (LWC) and ice water path (LWP) for ice clouds. Left panel shows the seasonal mean profiles of IWC. Right panel shows the monthly mean profiles of IWC and monthly box-plot of IWP. For the boxplots, medians are denoted by red lines and the average by black stars.

Our investigation into precipitating ice clouds highlights significant ice water content (IWC) between 2 and 6 km throughout the year, with values ranging from 100 to 700 mg m^{-3} . These clouds contain the largest amount of ice compared to other cloud types, with particularly high IWC during winter, summer, and early fall. In summer, IWC values are found at slightly higher altitudes, consistent with the median cloud base located at 4 km. Notably, July exhibits the highest IWC, reaching 1000 mg m^{-3} at 9 km, despite the minimal occurrence of these clouds, as observed from a 5-year dataset.

Boxplot analysis of liquid water path (LWP) indicates that IWC is relatively high in these clouds year-round due to their high thickness and small portion of liquid water. However, the substantially higher summer values create an impression of lower annual values. July, with significantly fewer occurrences than August, results in a higher average LWP in August. This pattern is also seen in September, where higher average LWP corresponds to fewer occurrences of high IWC clouds.

Seasonal profiles of IWC reveal distinct behaviors. Winter and spring exhibit similar vertical profiles with peaks at 4 km, with winter showing slightly higher values below this level, and nearly equal values above. Summer and fall also share similar profiles, both peaking at 6 km, though fall has higher values below this peak, and equal values above. Additionally, summer features a distinct peak at around 10 km, slightly higher than the maximum observed in July. This is due to averaging data over the entire summer, shifting the peak 1 km lower when considering the season as a whole.

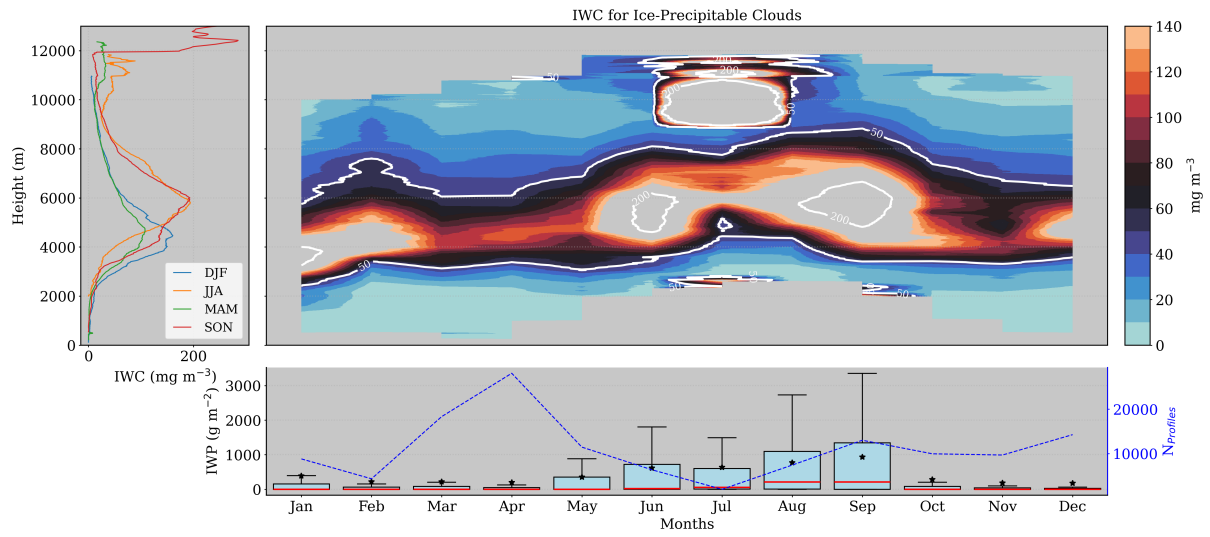


Figure 23. Same as Figure 22, but for mixed-phase clouds.

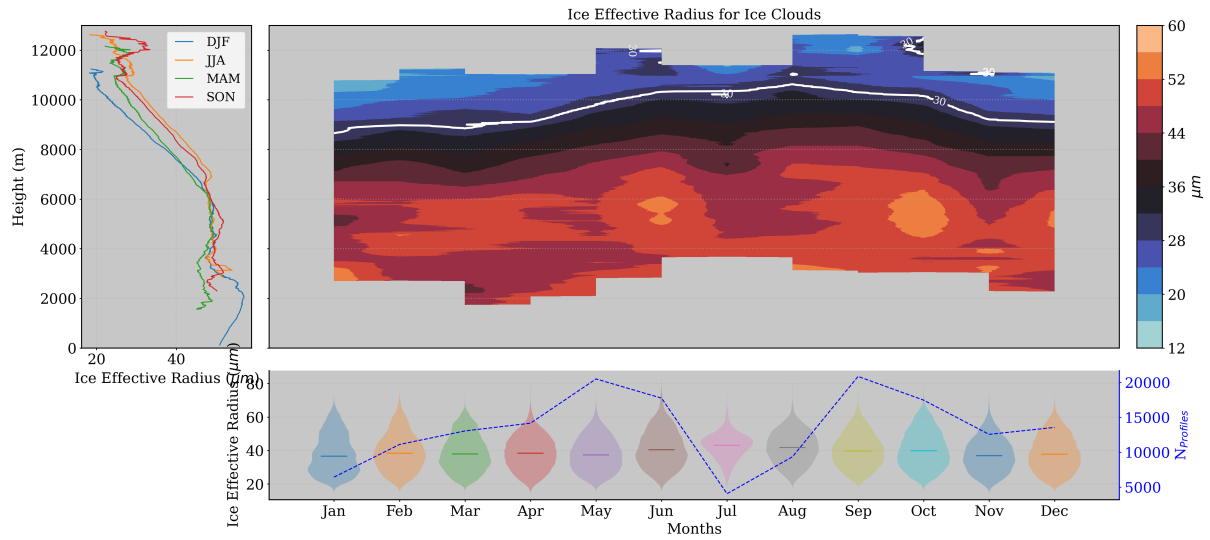


Figure 24. Seasonal and monthly average of ice water content (LWC) and ice water path (LWP) for ice clouds. Left panel shows the seasonal mean profiles of IWC. Right panel shows the monthly mean profiles of IWC and monthly box-plot of IWP. For the boxplots, medians are denoted by red lines and the average by black stars.

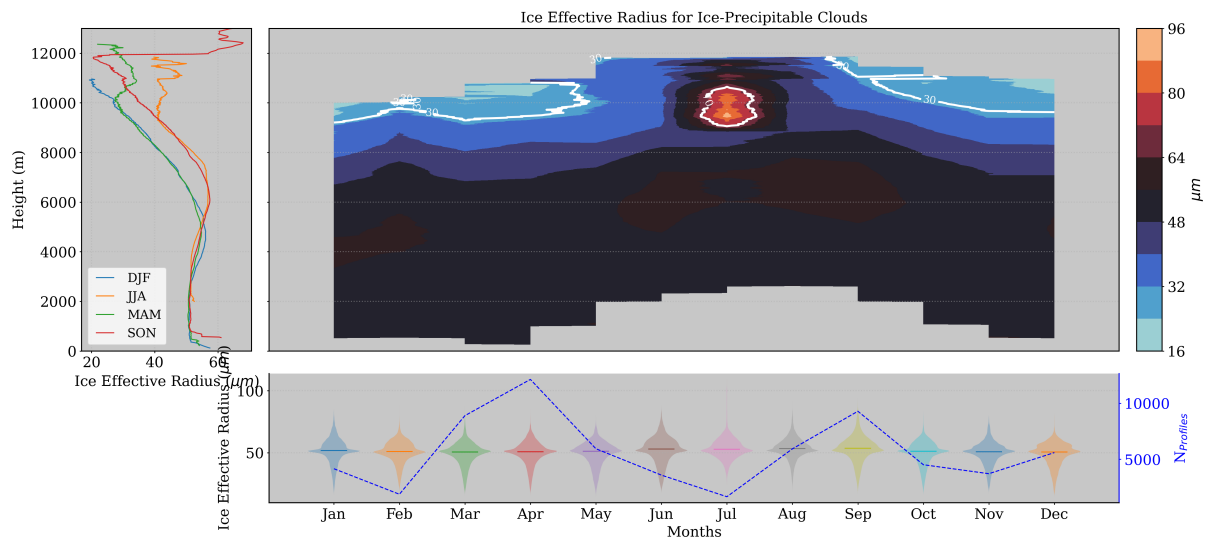


Figure 25. Same as Figure 22, but for mixed-phase clouds.

Acknowledgements. TEXT

References

- 465 A. Cazorla, Cazorla, A., Juan Andrés Casquero-Vera, Casquero-Vera, J. A., Roberto Román, Román, R., Juan Luís Guerrero-Rascado, Guerrero-Rascado, J. L., Carlos Toledano, Toledano, C., Victoria E. Cachorro, Cachorro, V. E., J.A.G. Orza, Orza, J., M. L. Cencillo, Cencillo, M. L., A. Serrano, Serrano, A., Gloria Titos, Titos, G., M. Pandolfi, Pandolfi, M., Andrés Alastuey, Alastuey, A., Natalie Hanrieder, Hanrieder, N., L. Alados-Arboledas, and Alados-Arboledas, L.: Near-Real-Time Processing of a Ceilometer Network Assisted with Sun-Photometer Data: Monitoring a Dust Outbreak over the Iberian Peninsula, 17, 11 861–11 876, [https://doi.org/10.5194/acp-17-](https://doi.org/10.5194/acp-17-11861-2017)
- 470 11861-2017.
- A. S. Frisch, Frisch, A. S., Graham Feingold, Feingold, G., C. W. Fairall, Fairall, C. W., Taneil Uttal, Uttal, T., J. B. Snider, J. B. Snider, Snider, J. B., and J. B. Snider: On Cloud Radar and Microwave Radiometer Measurements of Stratus Cloud Liquid Water Profiles, 103, 23 195–23 197, <https://doi.org/10.1029/98jd01827>.
- Alexandre Rios-Entenza, Rios-Entenza, A., Pedro M. M. Soares, Soares, P. M. M., Ricardo M. Trigo, Trigo, R. M., Rita M. Cardoso, Cardoso, R. M., Gonzalo Miguez-Macho, and Miguez-Macho, G.: Moisture Recycling in the Iberian Peninsula from a Regional Climate Simulation: Spatiotemporal Analysis and Impact on the Precipitation Regime, 119, 5895–5912, <https://doi.org/10.1002/2013jd021274>.
- 475 Casquero-Vera, J. A., Lyamani, H., Dada, L., Hakala, S., Paasonen, P., Román, R., Fraile, R., Petäjä, T., Olmo-Reyes, F. J., and Alados-Arboledas, L.: New Particle Formation at Urban and High-Altitude Remote Sites in the South-Eastern Iberian Peninsula, 20, 14 253–14 271, <https://doi.org/10.5194/acp-20-14253-2020>.
- 480 Enrique Pravia-Sarabia, Juan Pedro Montávez, Amar Halifa-Marín, Pedro Jiménez-Guerrero, and Juan José Gómez-Navarro: The Role of Aerosol Concentration on Precipitation in a Winter Extreme Mixed-Phase System: The Case of Storm Filomena, 15, 1398–1398, <https://doi.org/10.3390/rs15051398>.
- F. J. Bello-Millán, Julián Palacios, Paloma Gutierrez-Castillo, and Luis Parras: Simulations of a Tornadoic Supercell Event in the South of Spain: Sensitivity to Initial and Boundary Conditions and Microphysics Parameterizations, <https://doi.org/10.1016/j.atmosres.2024.107262>.
- 485 Ferreira, R. N.: Cut-Off Lows and Extreme Precipitation in Eastern Spain: Current and Future Climate, 12, 835, <https://doi.org/10.3390/atmos12070835>.
- Francisco Navas-Guzmán, Navas-Guzmán, F., J. Fernández-Gálvez, Fernández-Gálvez, J., Jesús Fernández-Gálvez, María José Granados-Muñoz, Granados-Muñoz, M. J., Juan Luís Guerrero-Rascado, Guerrero-Rascado, J. L., Juan Antonio Bravo-Aranda, Bravo-Aranda, J. A.,
- 490 L. Alados-Arboledas, and Alados-Arboledas, L.: Tropospheric Water Vapour and Relative Humidity Profiles from Lidar and Microwave Radiometry, 7, 1201–1211, <https://doi.org/10.5194/amt-7-1201-2014>.
- Hannes Griesche, Griesche, H., Patric Seifert, Seifert, P., Albert Ansmann, Ansmann, A., Holger Baars, Baars, H., Carola Barrientos Velasco, Velasco, C. B., Johannes Bühl, Bühl, J., Ronny Engelmann, Engelmann, R., Martin Radenz, Radenz, M., Yin, Z., Zhenping Yin, Zhenping, Y., Andreas Macke, and Macke, A.: Application of the Shipborne Remote Sensing Supersite OCEANET for Profiling of Arctic Aerosols and Clouds during Polarstern Cruise PS106, 13, 5335–5358, <https://doi.org/10.5194/amt-2019-434>.
- 495 Hogan, R. J. and O'Connor, E. J.: Facilitating Cloud Radar and Lidar Algorithms: The Cloudnet Instrument Synergy/Target Categorization Product.
- J. A., Illingworth, J. R., Hogan, ' E.J.o, Connor, D., Bouniol, E. M., Brooks, J., Delanoë, P. D., Donovan, D. J., Eastment, N., Gaussiat, F. J.W., Goddard, M., Haefelin, H., Klein Baltink, A. O., Krasnov, Pelon, J.-M., Piriou, A., Protat, J. H.W., Russchenberg, Seifert, M. A.,

- 500 Tompkins, G.-J., V. Zadelhoff, F., Vinit, U., Willén, R. D., and Wilson: Continuous Evaluation of Cloud Profiles in Seven Operational Models Using Ground-Based Observations.
- Julien Delanoë, Delanoë, J., Alain Protat, Protat, A., Alain Protat, Dominique Bouniol, Bouniol, D., Bouniol, D., Andrew J. Heymsfield, Heymsfield, A. J., Aaron Bansemer, Bansemer, A., Philip R. A. Brown, Brown, P., and Brown, P. R. A.: The Characterization of Ice Cloud Properties from Doppler Radar Measurements, 46, 1682–1698, <https://doi.org/10.1175/jam2543.1>.
- 505 Marcos García Sotillo, Sotillo, M. G., Baptiste Mourre, Mourre, B., Marc Mestres, Mestres, M., Pablo Lorente, Lorente, P., Roland Aznar, Roland Aznar, Aznar, R., Manuel García-León, García-León, M., María Liste, Maria Liste, Maria Liste, Liste, M., Ana Lúcia Almeida Santana, Santana, A., Manuel Espino, Espino, M., Enrique Álvarez, and Álvarez, E.: Evaluation of the Operational CMEMS and Coastal Downstream Ocean Forecasting Services During the Storm Gloria (January 2020), 8, <https://doi.org/10.3389/fmars.2021.644525>.
- Marinou, E., Voudouri, K. A., Tsikoudi, I., Drakaki, E., Tsekeri, A., Rosoldi, M., Ene, D., Baars, H., O'Connor, E., Amiridis, V., and Meleti, C.: Geometrical and Microphysical Properties of Clouds Formed in the Presence of Dust above the Eastern Mediterranean, 13, 5001, <https://doi.org/10.3390/rs13245001>.
- 510 Marta de Alfonso, family=Alfonso, given=Marta, p. u., Jue Lin-Ye, Lin-Ye, J., José María García-Valdecasas, Garcia-Valdecasas, J. M., Susana Pérez-Rubio, Pérez-Rubio, S., M. Y. Luna, M. Yolanda Luna, Luna, M. Y., Daniel Santos-Muñoz, Santos-Muñoz, D., María Ángeles Martínez Ruiz, Ruiz, M. I., Begoña Pérez-Gómez, Pérez-Gómez, B., Enrique Álvarez-Fanjul, and Álvarez Fanjul, E.: Storm Gloria: Sea State Evolution Based on in Situ Measurements and Modeled Data and Its Impact on Extreme Values, 8, <https://doi.org/10.3389/fmars.2021.646873>.
- 515 Nils Küchler, Küchler, N., Stefan Kneifel, Kneifel, S., Ulrich Löhnert, Löhnert, U., Pavlos Kollias, Kollias, P., Harald Czekala, Czekala, H., Thomas Rose, and Rose, T.: A W-Band Radar–Radiometer System for Accurate and Continuous Monitoring of Clouds and Precipitation, 34, 2375–2392, <https://doi.org/10.1175/jtech-d-17-0019.1>.
- 520 Pablo Saavedra Garfias, Heike Kalesse-Los, Luisa von Albedyll, Hannes Griesche, and G. Spreen: Asymmetries in Cloud Microphysical Properties Ascribed to Sea Ice Leads via Water Vapour Transport in the Central Arctic, <https://doi.org/10.5194/acp-23-14521-2023>.
- Porcù, F., Carrassi, A., Medaglia, C. M., Prodi, F., and Mugnai, A.: A Study on Cut-off Low Vertical Structure and Precipitation in the Mediterranean Region, 96, 121–140, <https://doi.org/10.1007/s00703-006-0224-5>.
- Robin J. Hogan, Hogan, R. J., Marion Mittermaier, Mittermaier, M. P., Anthony J. Illingworth, and Illingworth, A. J.: The Retrieval of Ice Water Content from Radar Reflectivity Factor and Temperature and Its Use in Evaluating a Mesoscale Model, 45, 301–317, <https://doi.org/10.1175/jam2340.1>.
- 525 Răzvan Pîrloagă, Răzvan Pîrloagă, Ene, D., Dragoș Ene, Mihai Boldeanu, Mihai Boldeanu, Bogdan Antonescu, Bogdan Antonescu, Ewan J. O'Connor, Ewan O'Connor, Sabina Ștefan, and Sabina Ștefan: Ground-Based Measurements of Cloud Properties at the Bucharest–Măgurele Cloudnet Station: First Results, 13, 1445–1445, <https://doi.org/10.3390/atmos13091445>.
- 530 S. Frisch, Frisch, S., Matthew D. Shupe, Shupe, M. D., Irina V. Djalalova, Djalalova, I., Graham Feingold, Feingold, G., Michael R. Poellot, and Poellot, M. R.: The Retrieval of Stratus Cloud Droplet Effective Radius with Cloud Radars, 19, 835–842, [https://doi.org/10.1175/1520-0426\(2002\)019<0835:trosed>2.0.co;2](https://doi.org/10.1175/1520-0426(2002)019<0835:trosed>2.0.co;2).
- Shahi, N. K., Polcher, J., Bastin, S., Pennel, R., and Fita, L.: Assessment of the Spatio-Temporal Variability of the Added Value on Precipitation of Convection-Permitting Simulation over the Iberian Peninsula Using the RegIPSL Regional Earth System Model, 59, 471–498, <https://doi.org/10.1007/s00382-022-06138-y>.
- 535 Stefan Kneifel, Stefan Kneifel, Bernhard Pospichal, Bernhard Pospichal, Leonie von Terzi, Leonie von Terzi, Tobias Zinner, Tobias Zinner, Matjaž Puh, Matjaž Puh, Martin Hagen, Martin Hagen, Bernhard Mayer, Bernhard Mayer, Ulrich Löhnert, Ulrich Löhnert, Susanne

- Crewell, and Susanne Crewell: Multi-Year Cloud and Precipitation Statistics Observed with Remote Sensors at the High-Altitude Environmental Research Station Schneefernerhaus in the German Alps, <https://doi.org/10.1127/metz/2021/1099>.
- 540 Tatiana Nomokonova, Nomokonova, T., Kerstin Ebell, Ebell, K., Ulrich Löhnert, Löhnert, U., Marion Maturilli, Maturilli, M., Christoph Ritter, Ritter, C., Ewan O'Connor, and O'Connor, E.: Statistics on Clouds and Their Relation to Thermodynamic Conditions at Ny-Ålesund Using Ground-Based Sensor Synergy, 19, 4105–4126, <https://doi.org/10.5194/acp-19-4105-2019>.